

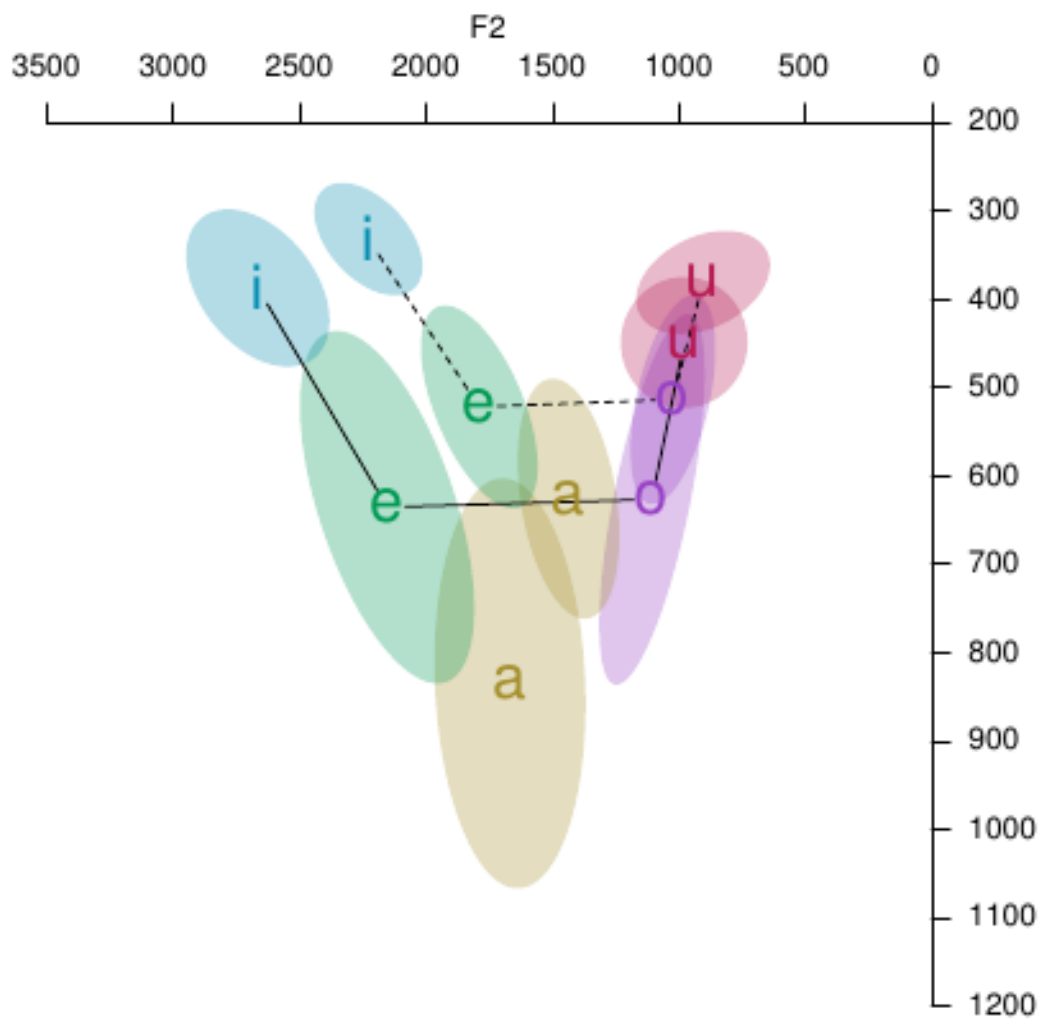
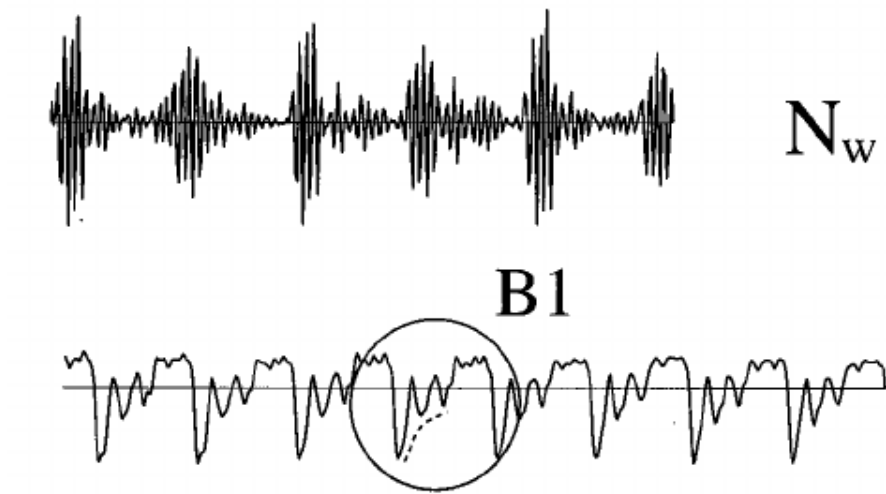
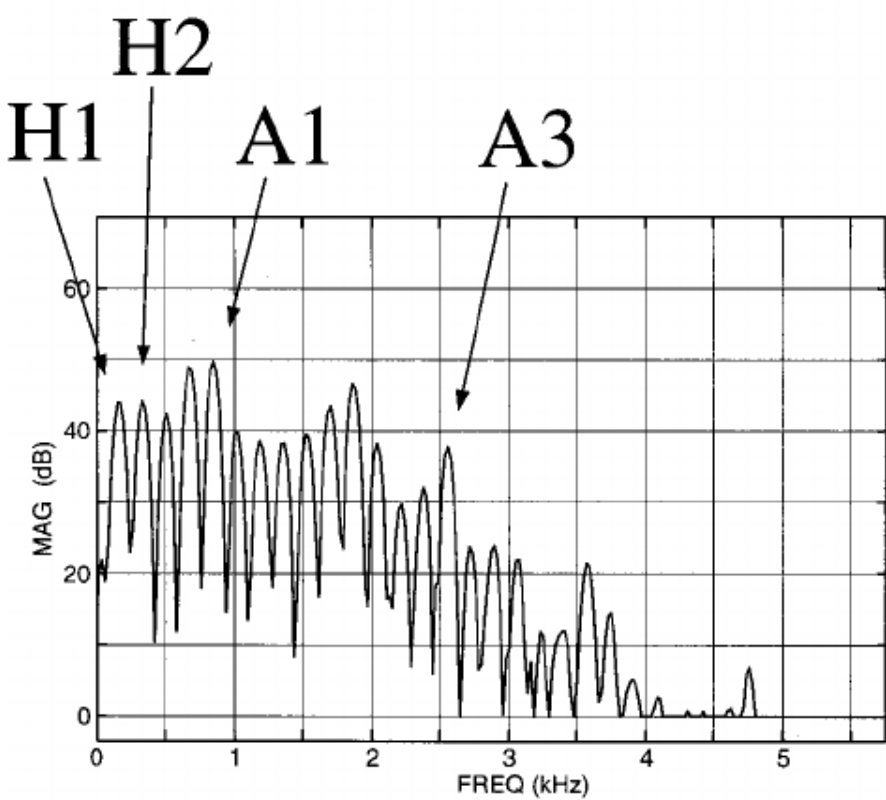
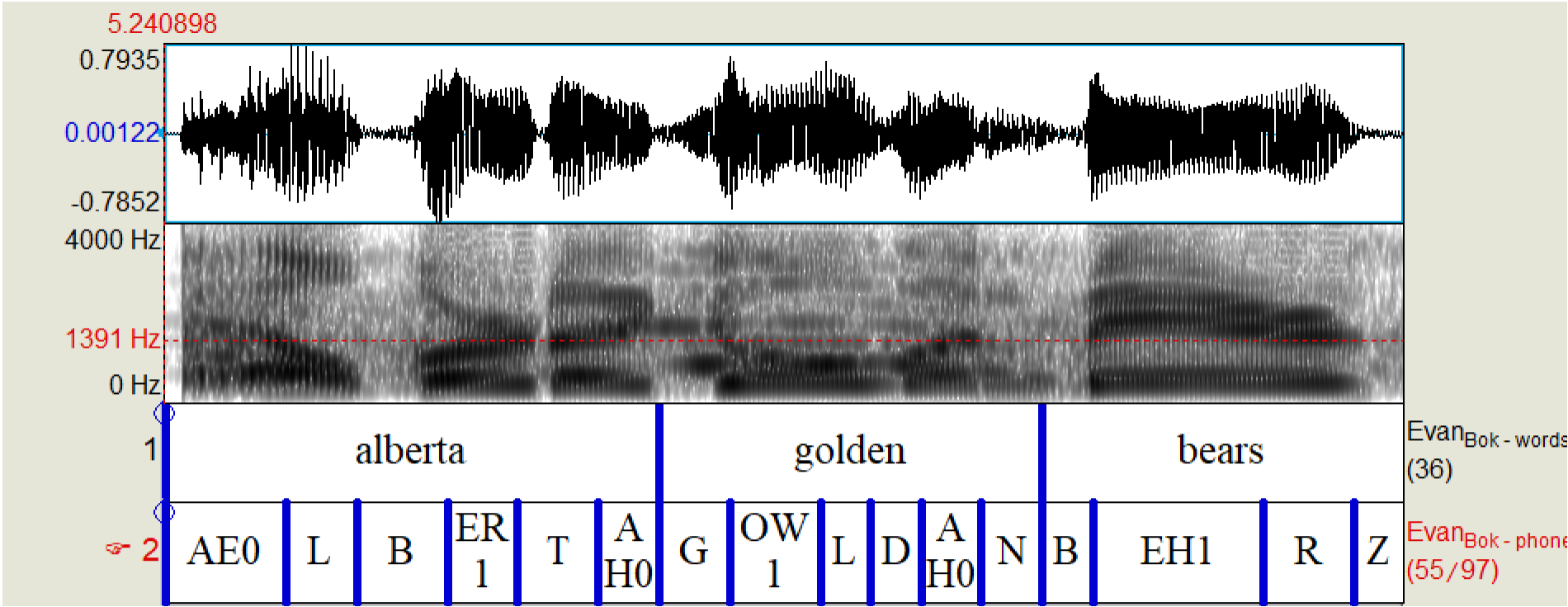
PolyglotDB

A library for representing and analysing speech data

James Tanner 2025/11/3 International Christian University

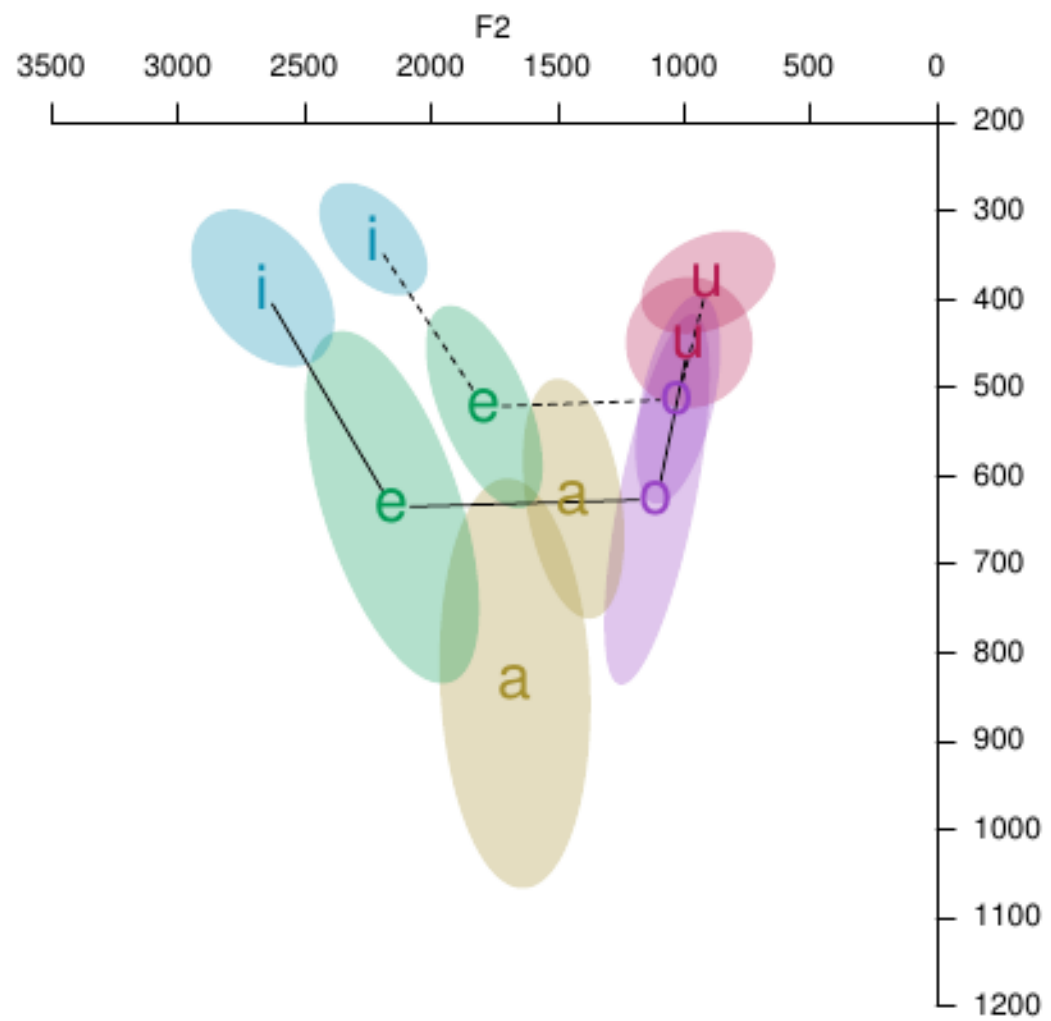
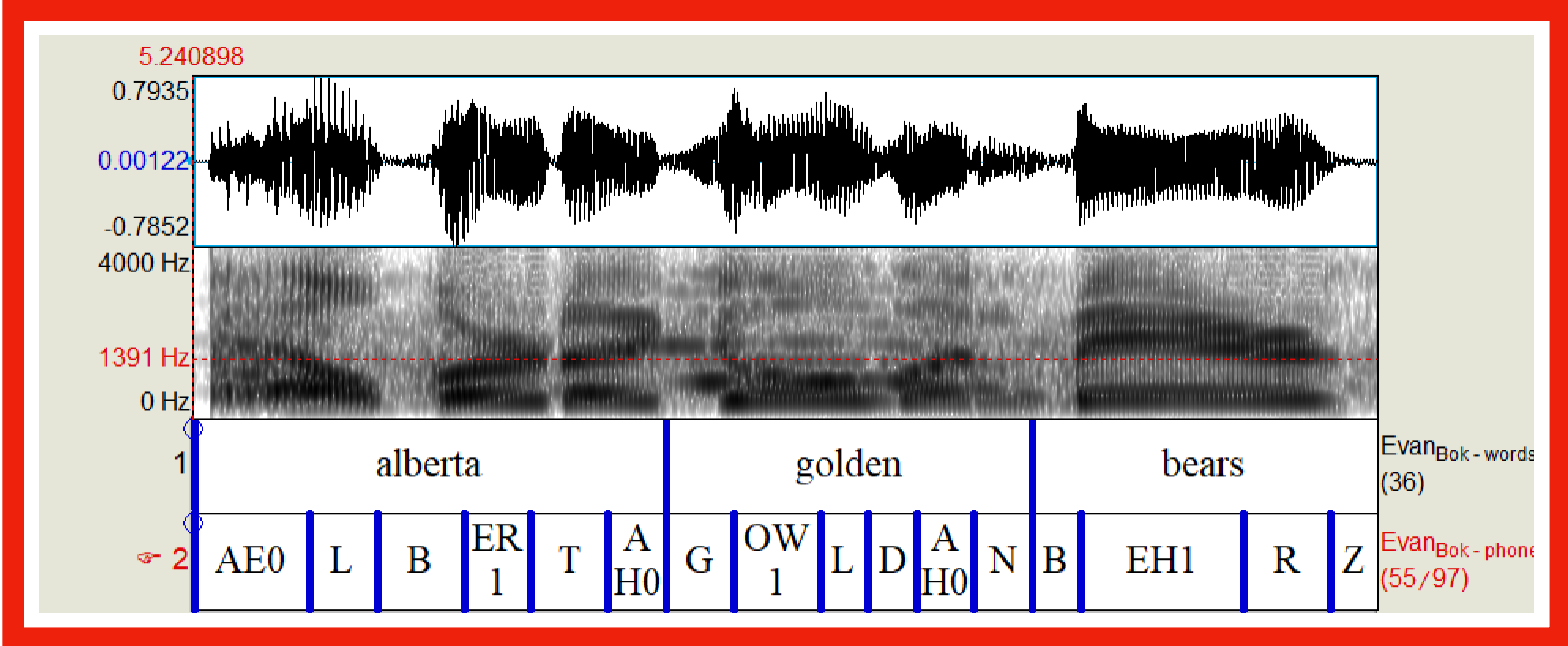


Corpus phonetics pipeline



1	a	a	6.68209
2	aaron	'e:ran	3.304921
3	aback	a'bak	2.465383
4	abacus	'abəkəs	1.690196
5	abandon	a'band(a)n	3.099681
6	abandoned	a'band(a)nd	3.532754
7	abandonment	a'bandanm(a)nt	2.158362
8	abase	a'bes	0.60206
9	abatement	a'besm(a)nt	0.60206
10	abash	a'baʃ	0
11	abate	a'best	1.792392
12	abbess	'abɛs	1.477121
13	abbey	'abi	3.576111
14	abbot	'abat	2.506505
15	abbreviate	a'brɪ:vɪət	1.431364
16	abbreviated	a'brɪ:vɪətɪd	2.352183
17	abbreviation	abrɪ:vɪ'etʃ(a)n	2.827369
18	abdicate	'abdɪkeɪt	1.672098
19	abdomen	'abdəməɪn	2.746634
20	abdominal	ab'dɒmɪn(a)l	2.522444
21	abduct	ab'dʌkt	1.740363
22	abduction	ab'dʌkʃn	2.453318
23	abed	a'bed	1.50515
24	abel	'ɛzb(a)l	2.257679
25	aberrant	a'ber(a)nt	0.69897
26	aberration	'abə'reɪʃ(a)n	1.857332
27	abet	a'bet	0.845098
28	abeyance	a'besəns	1.176091
29	abhor	ab'hɔ:	1.69897
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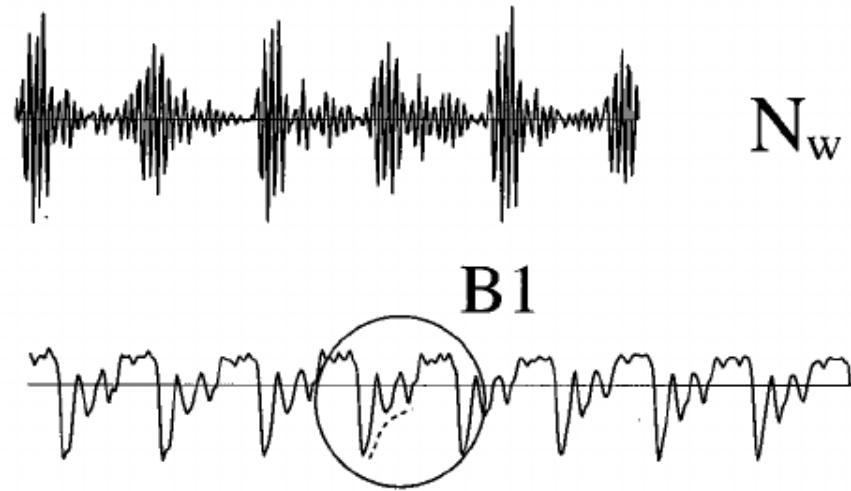
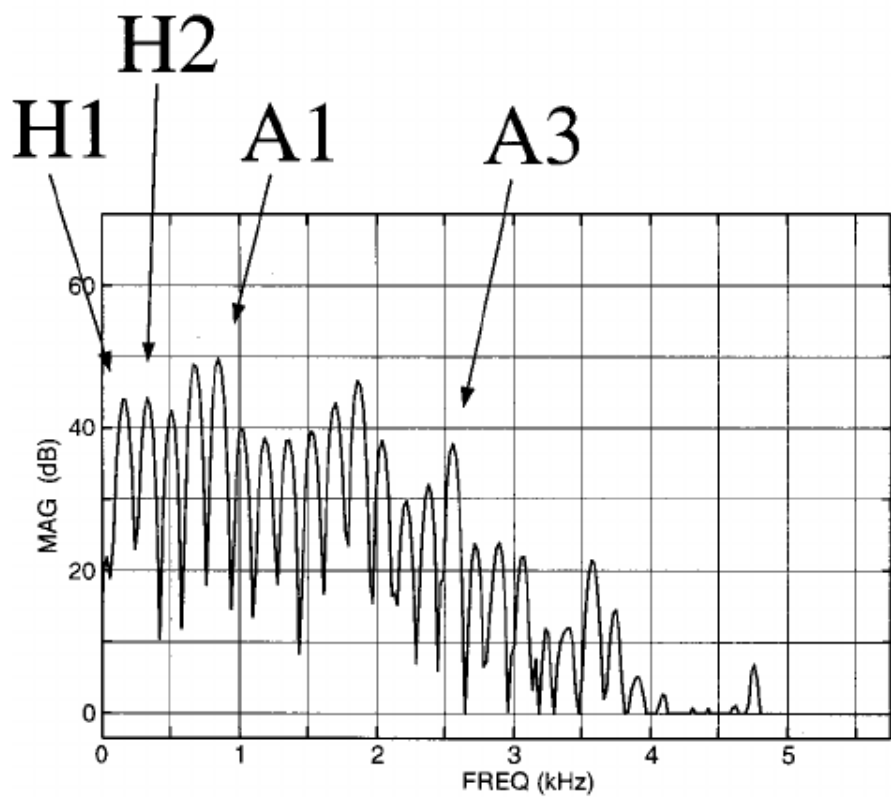
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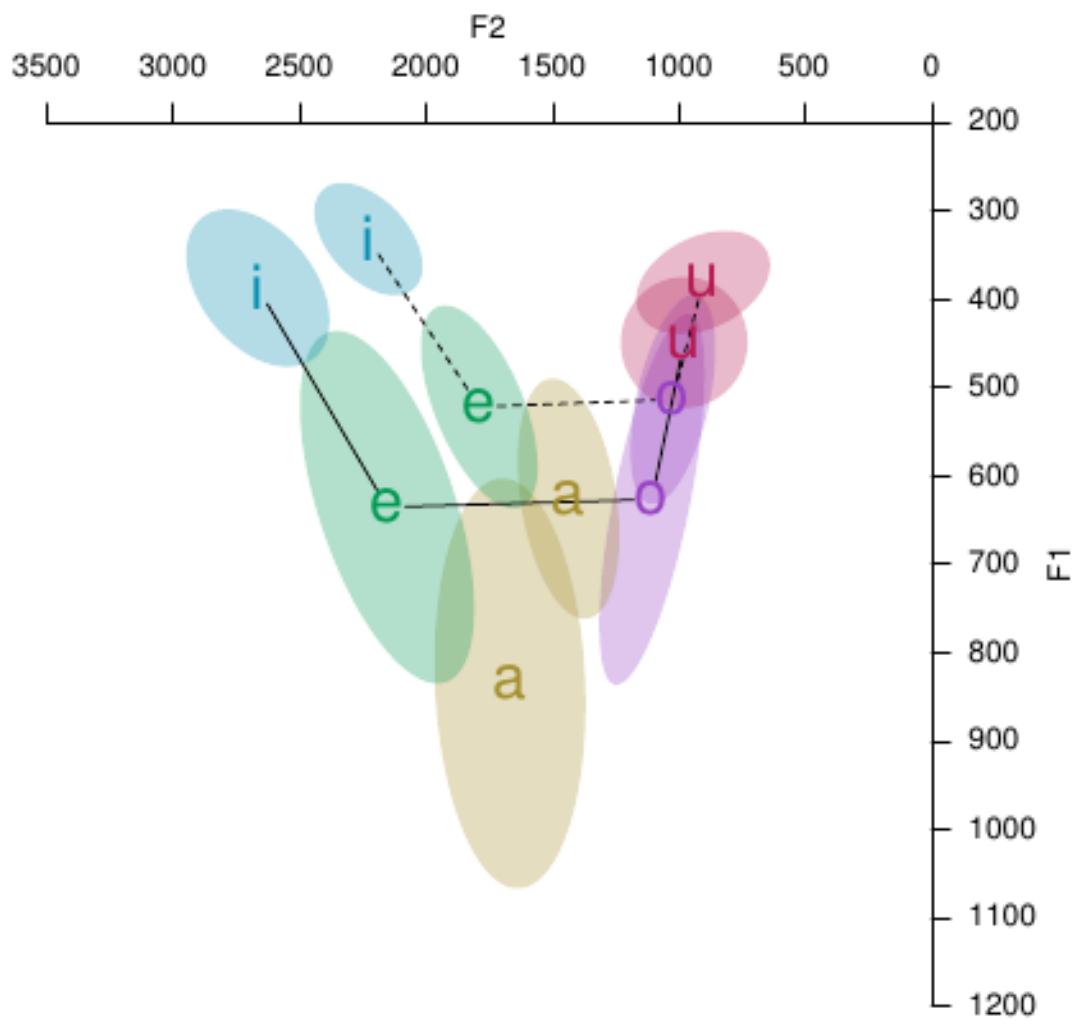
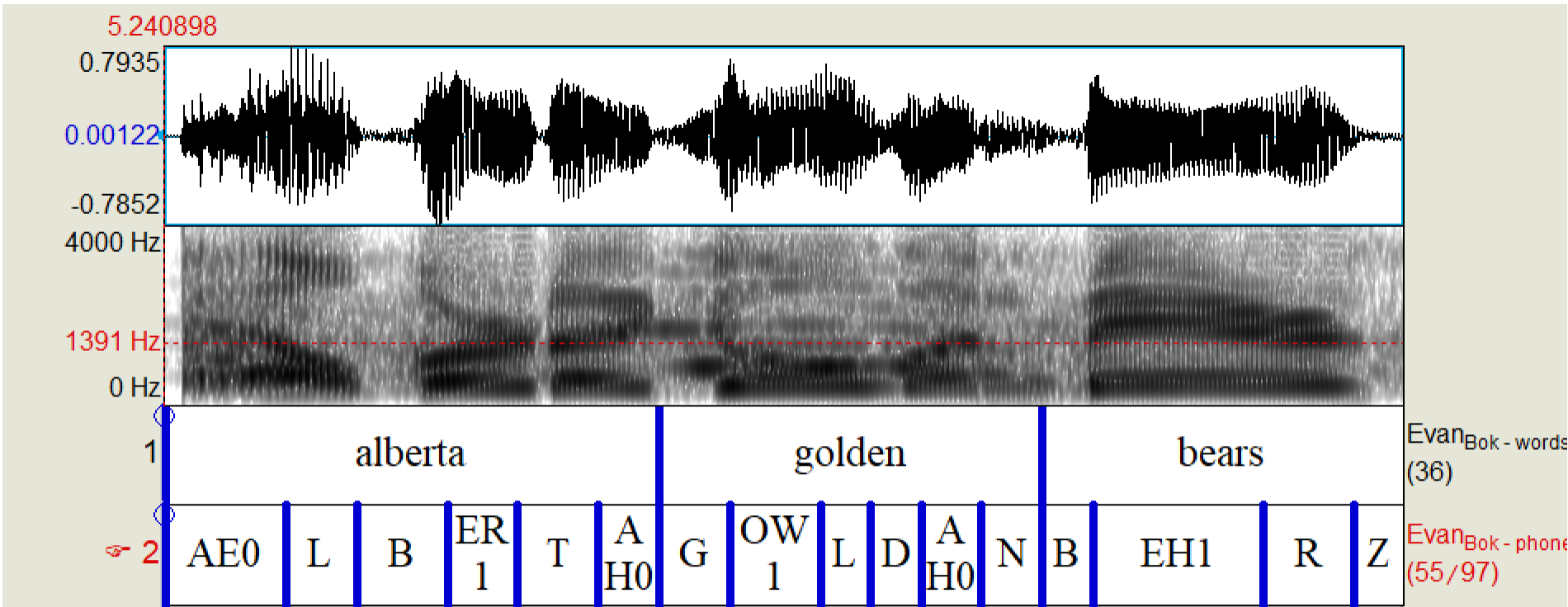
Last week

Speech-phone alignment with
Montreal Forced Aligner

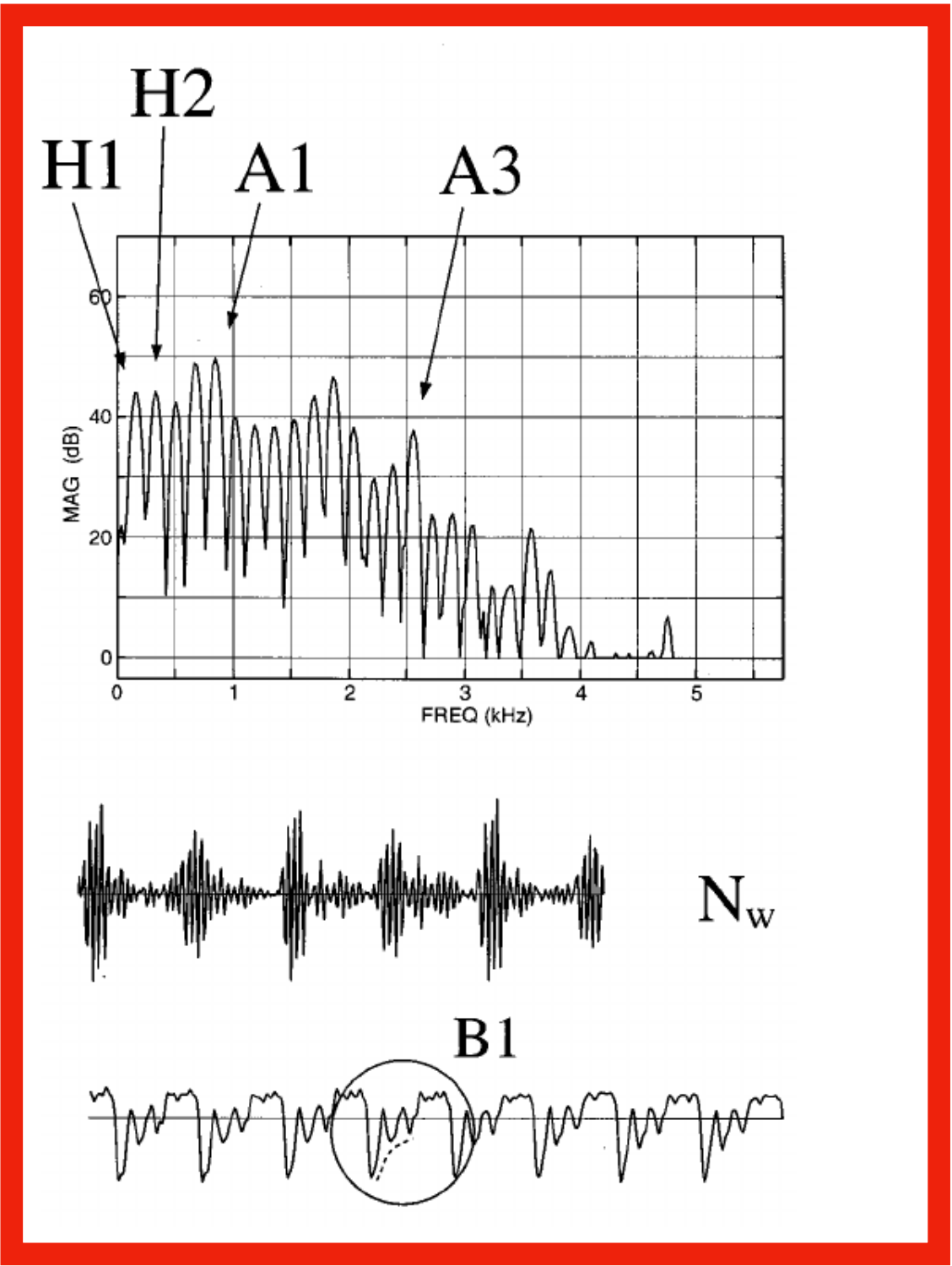


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Phonetic analysis and data extraction with PolyglotDB

Steps

- Typically involves
 - **Importing** data

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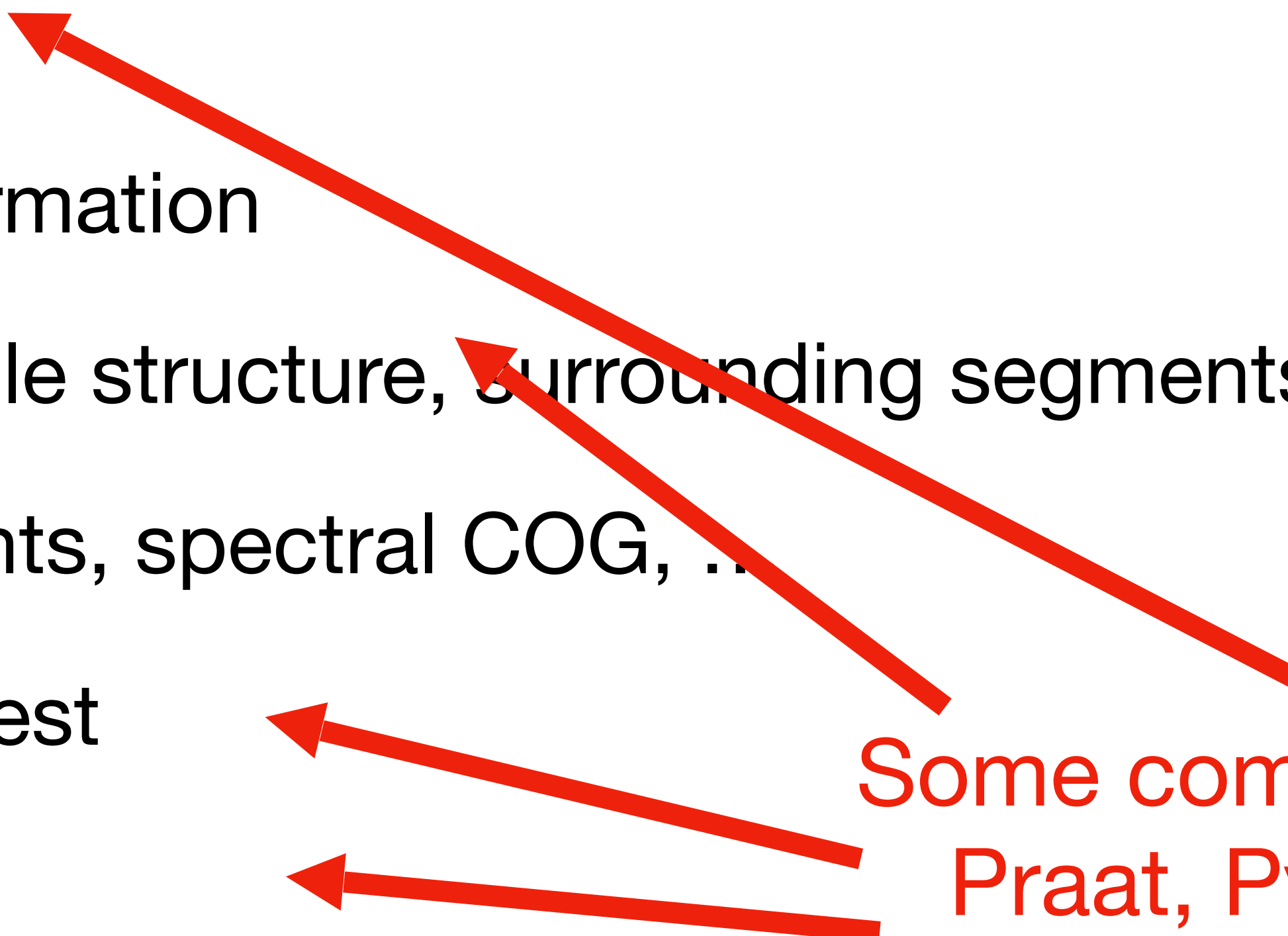
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Some combination of
Praat, Python, R,
VoiceSauce...



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Often written specifically
for each study & modified
for new cases

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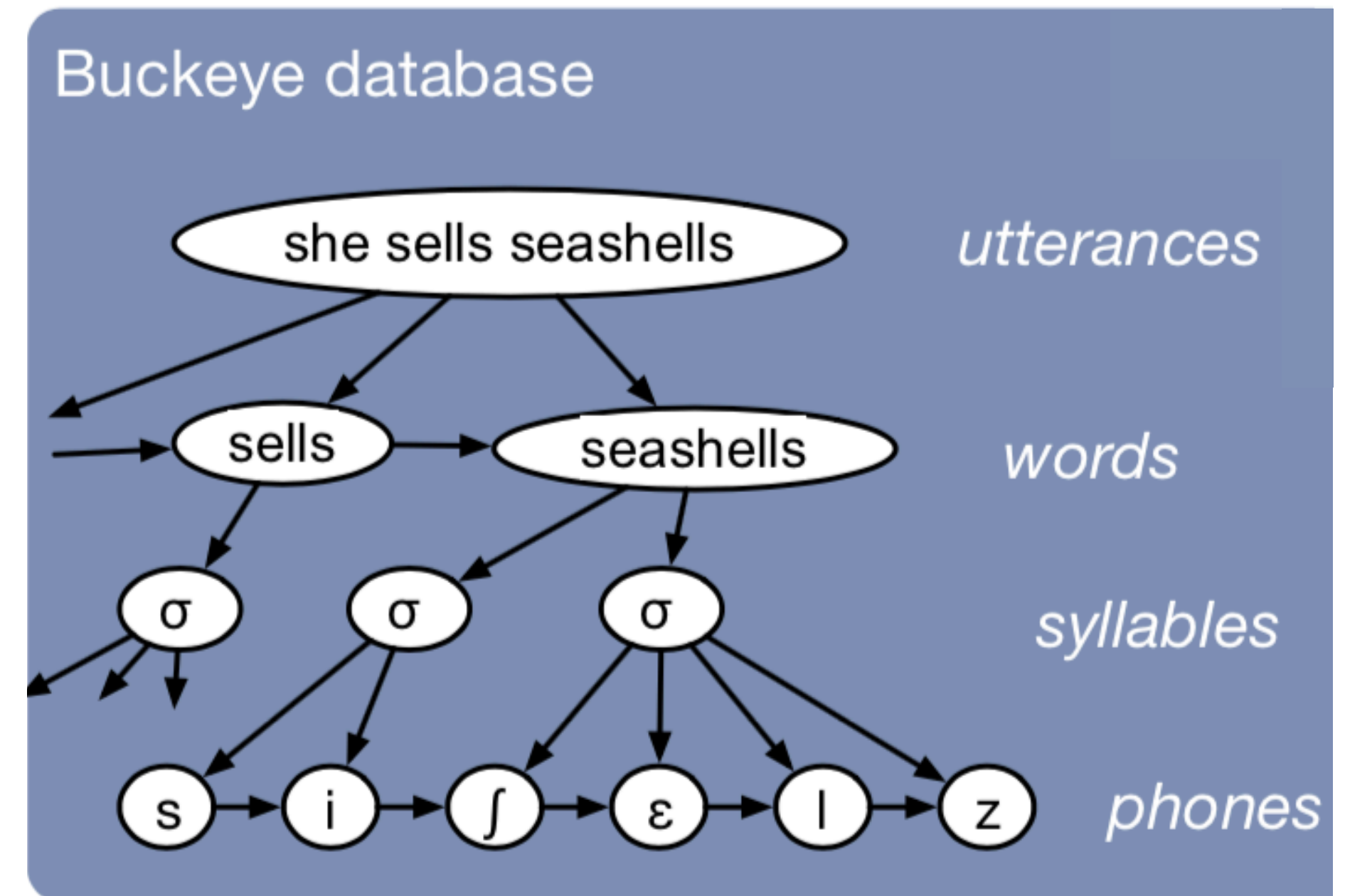
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Is there an alternative?

Speech database management systems

- Treat corpus as a **database**
 - Components are stored in structural relationship to each other
 - Use standardised methods & interface for making acoustic measurements, incorporating linguistic/speaker information, etc
 - Standardised querying and export mechanisms
 - (Sometimes) also include transcription/alignment functionality



Speech database management systems

System	Language	Features	Focus
CSJ-RDB	SQL	No reliance on custom software	Single corpus only
EMU-SDMS	R	Full alignment/ annotation/analysis intergration	Speech -> analysis pipeline
LaBB-Cat	Java		
PolyglotDB	Python	Handles multiple corpus formats Integration with Praat	Reproducible analyses across heterogenous formats

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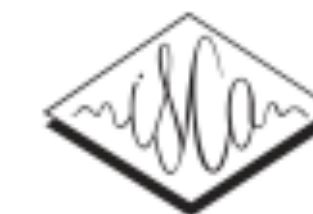
Today

PolyglotDB

- Python library for **representing, analysing, and querying** aligned speech data

INTERSPEECH 2017

August 20–24, 2017, Stockholm, Sweden



Polyglot and Speech Corpus Tools: a system for representing, integrating, and querying speech corpora

Michael McAuliffe¹, Elias Stengel-Eskin¹, Michaela Socolof², Morgan Sonderegger¹

¹Department of Linguistics, McGill University, Canada

²Department of Linguistics, University of Maryland, USA

ISCAN: A SYSTEM FOR INTEGRATED PHONETIC ANALYSES ACROSS SPEECH CORPORA

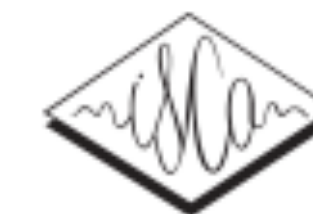
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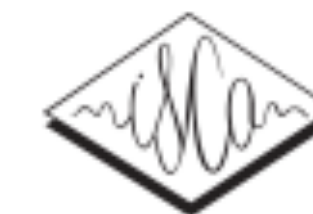
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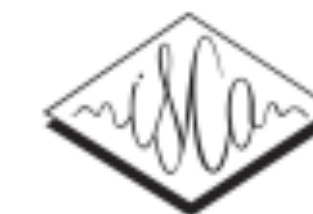
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PolyglotDB

- Python library for **representing, analysing, and querying** aligned speech data
- Requires **minimum** technical experience from the user
- **Abstracts away** from specific dataset format
- Methods for making linguistic and acoustic measurements both **standardised** and **customisable**

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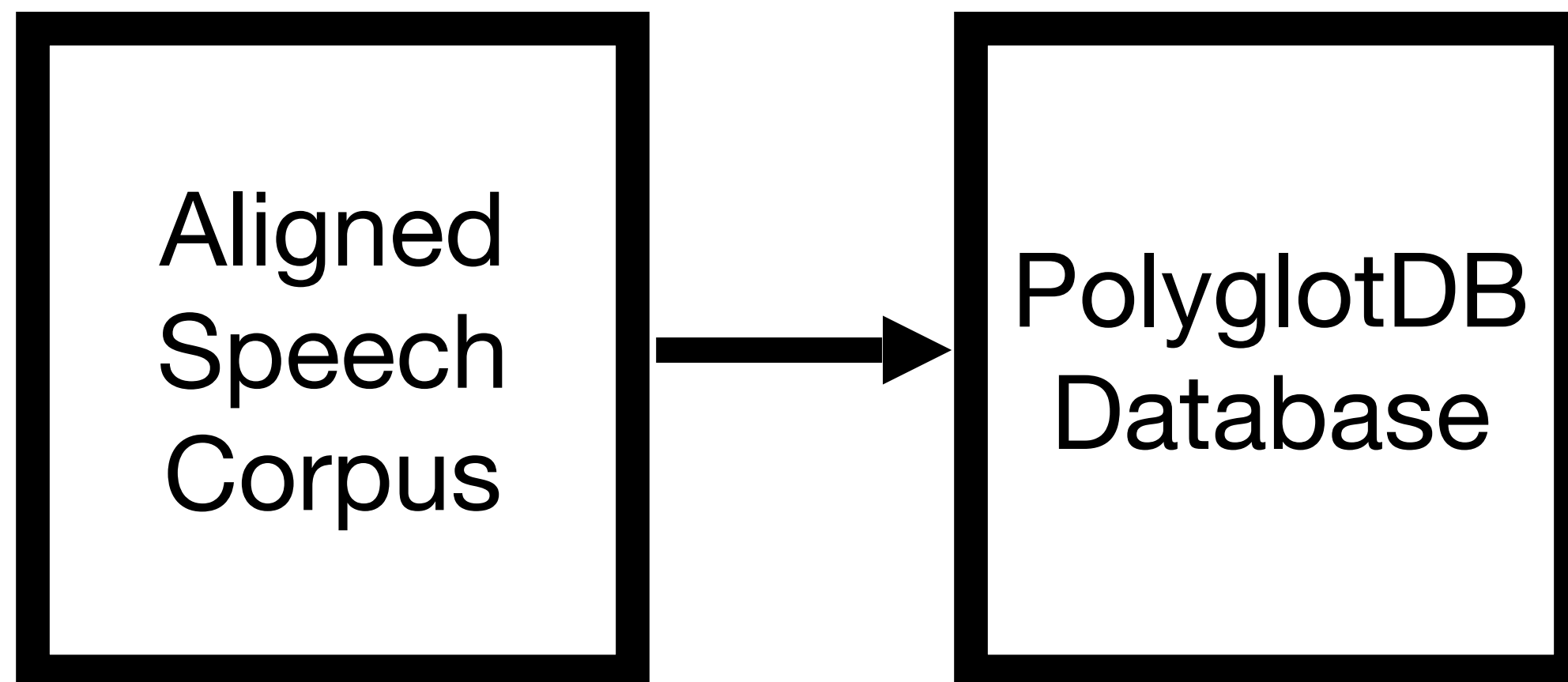
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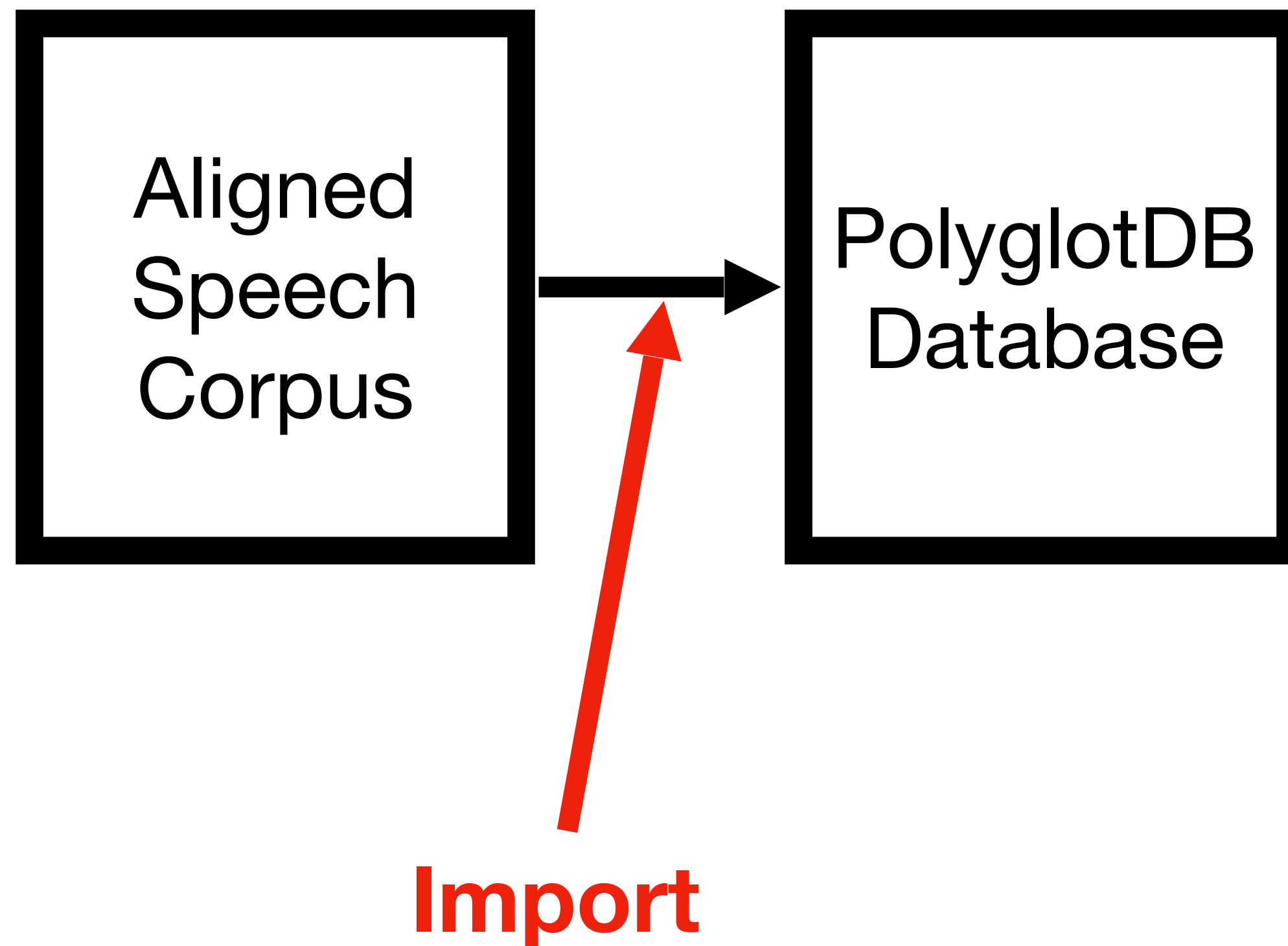
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PolyglotDB workflow

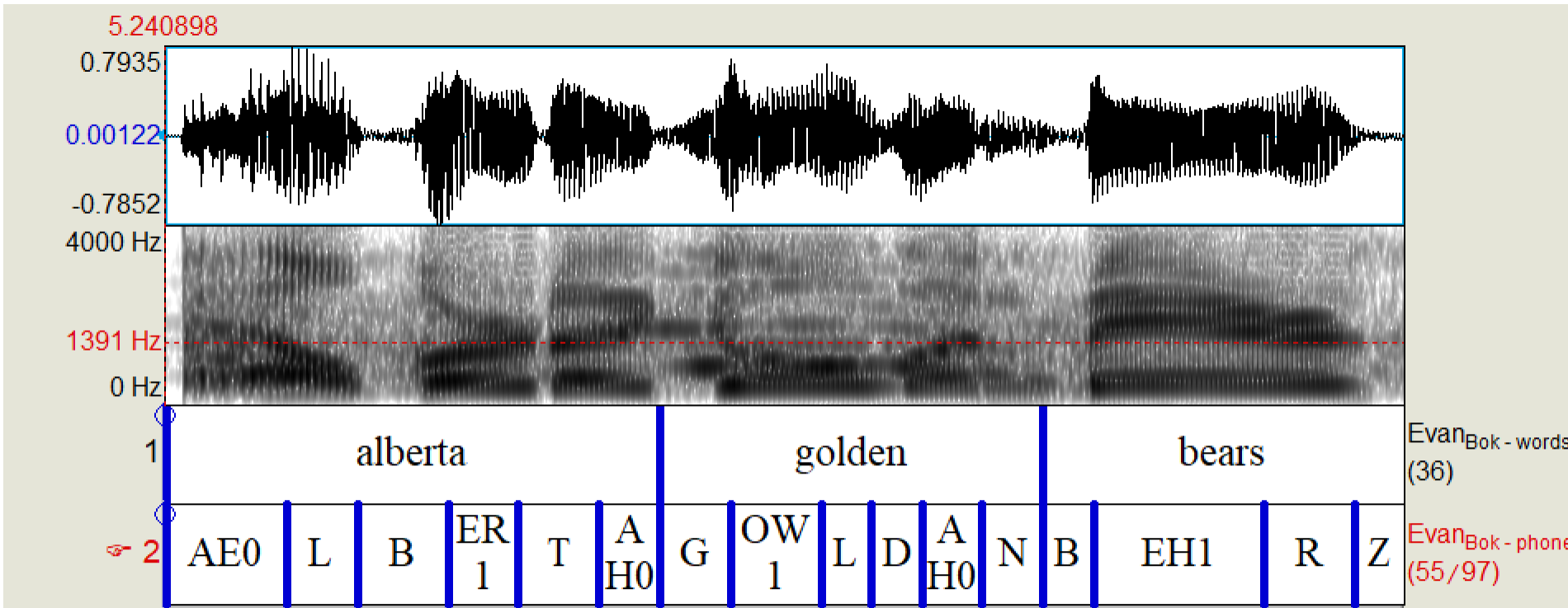


PolyglotDB workflow



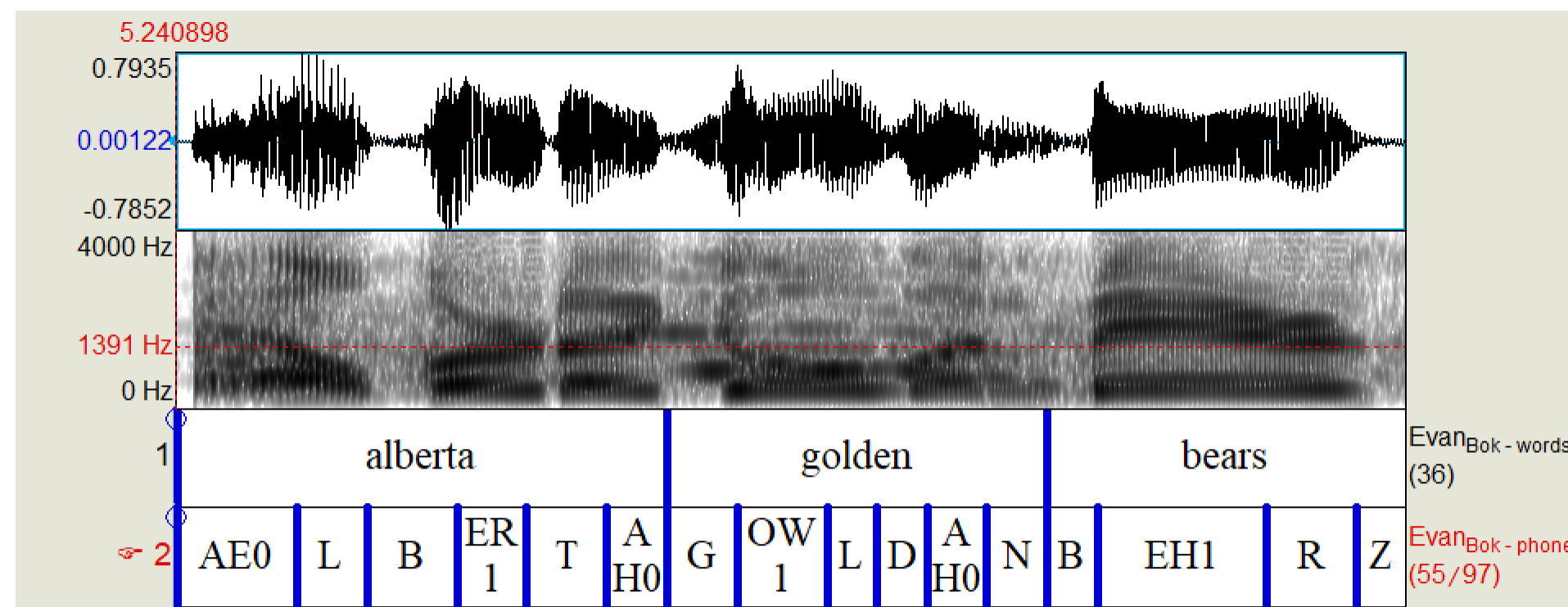
Import

Import of word and phone-align TextGrids



Import

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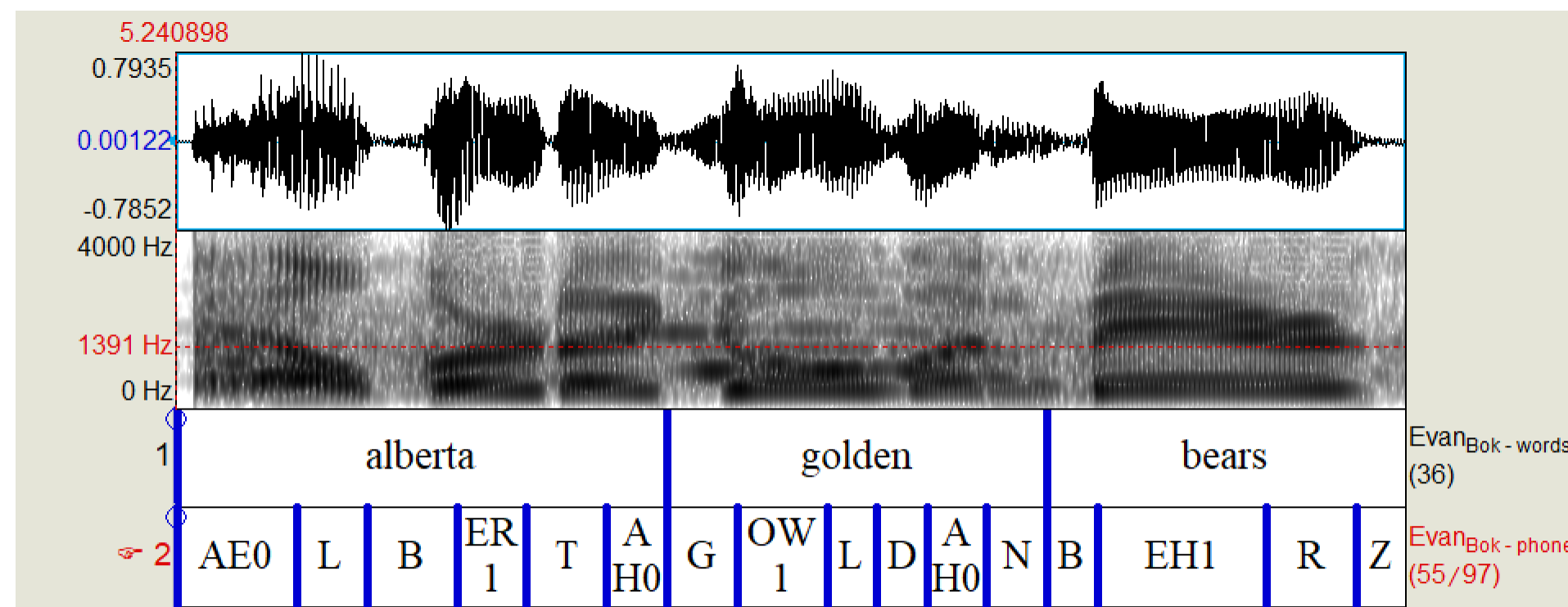
Supports:

- MFA, LaBB-Cat, FAVE, MAUS aligners
- Idiosyncratic formats (e.g. Buckeye, TIMIT)
- Julius support soon

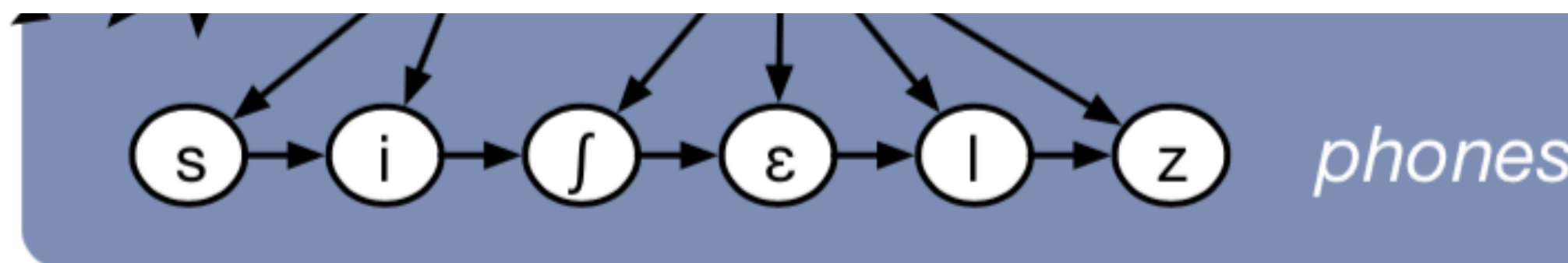
Possible to create your own custom corpus importers

Import

Import of word and phone-align
TextGrids



Database stores both **sequential** and **hierarchical** information about linguistic units



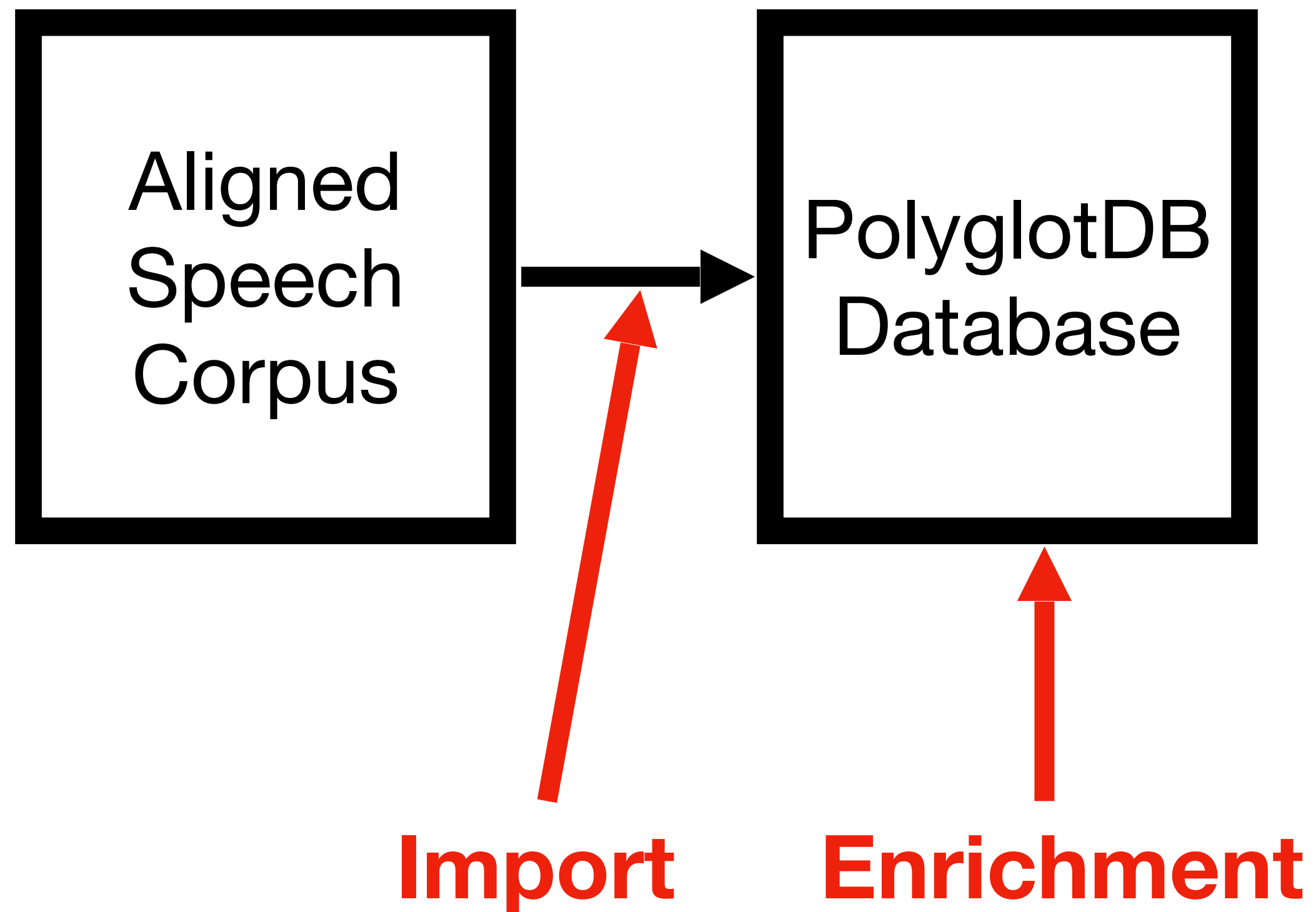
Each phone
belongs to a
word & speaker

Supports:

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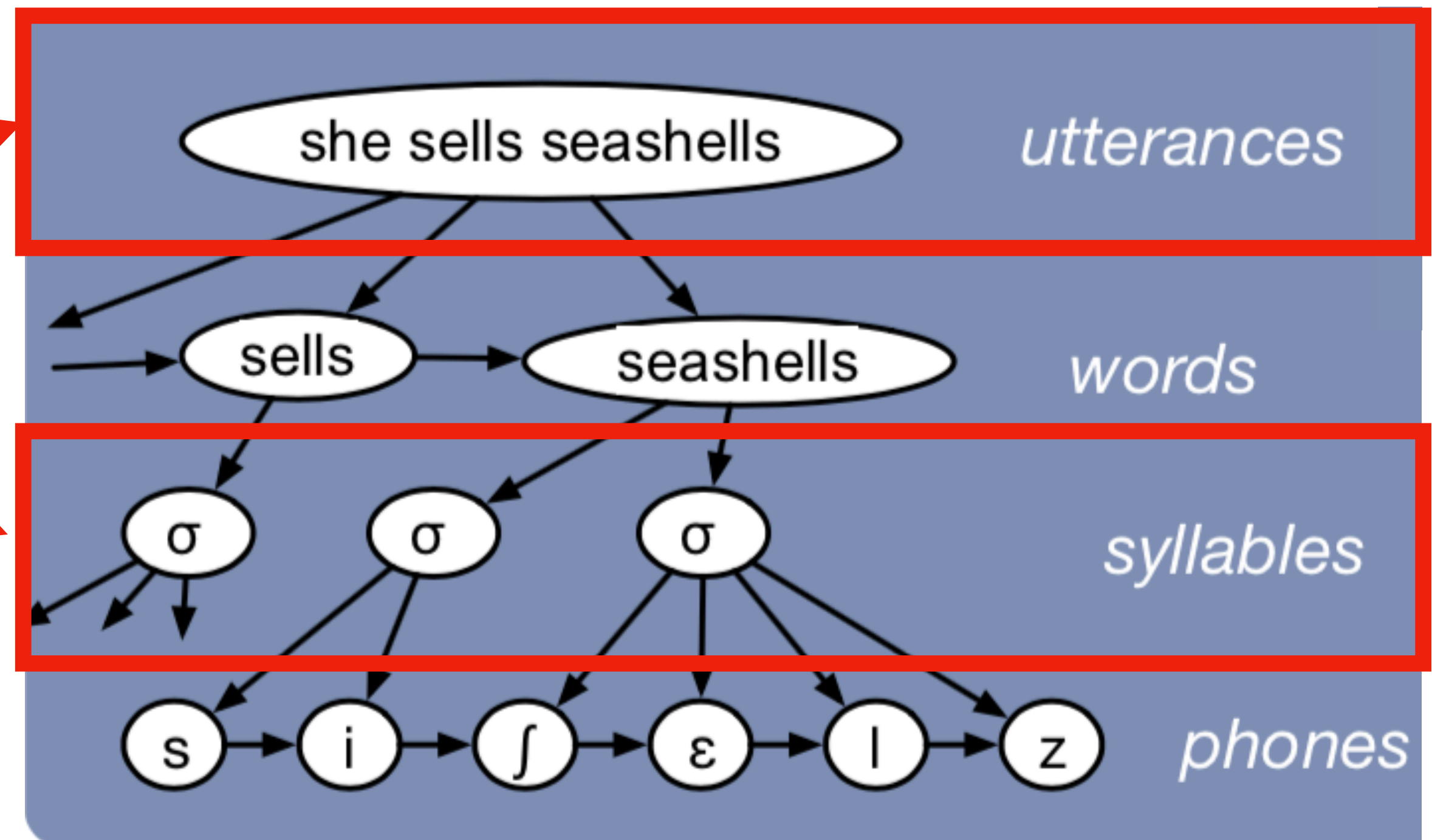
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PolyglotDB workflow



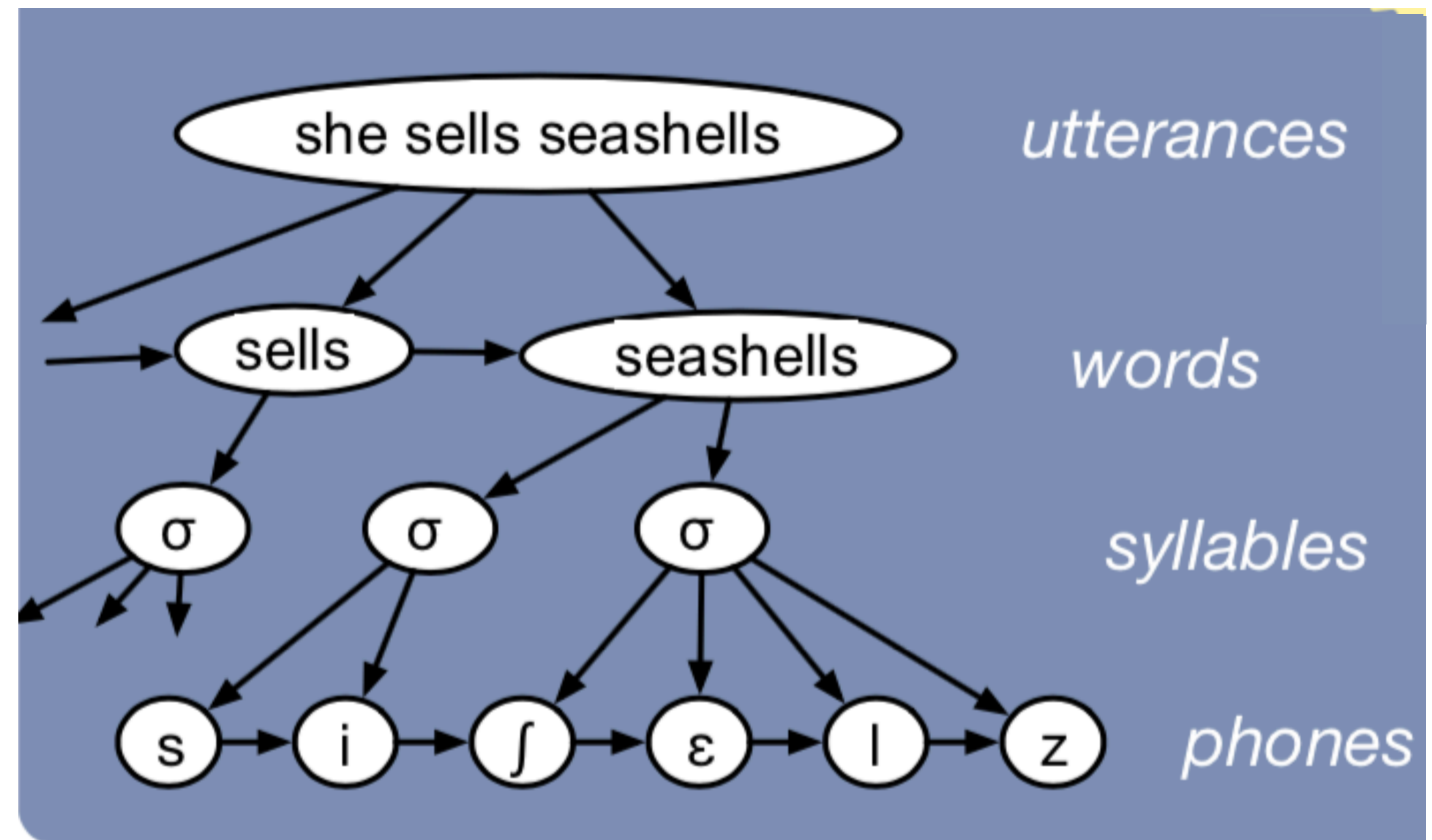
Enrichment

- Adding linguistic and acoustic information to the database
- Units
 - Pauses, utterances, syllables
- Measures:
 - Speech rate, stress



Enrichment

- Adding linguistic and acoustic information to the database
 - Units
 - Pauses, utterances, syllables
 - Measures:
 - Speech rate, stress
 - Lexical information:
 - Stress pattern, frequency
 - Speaker information:
 - Speaker gender, age, dialect



**From external lexicons/
metadata files**

Enrichment

- Acoustic measures

- Pitch/F0

- Vowel formants (single points & dynamic tracks)

- Intensity

**Internal methods
(Praat)**



Enrichment

- Acoustic measures

- Pitch/F0

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- VOT

**Internal methods
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AutoVOT

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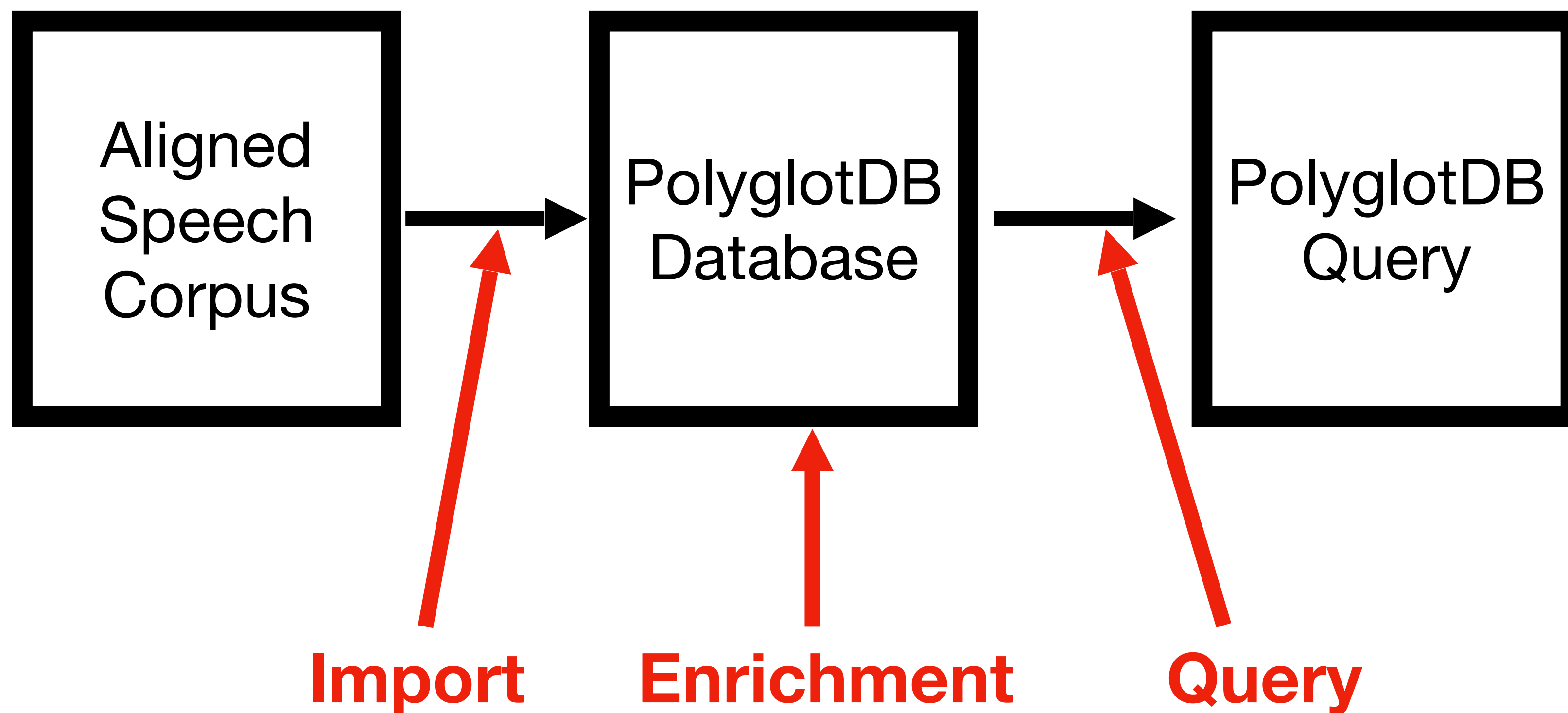
- Anything you want!

**Internal methods
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AutoVOT

**Custom Praat script
support**

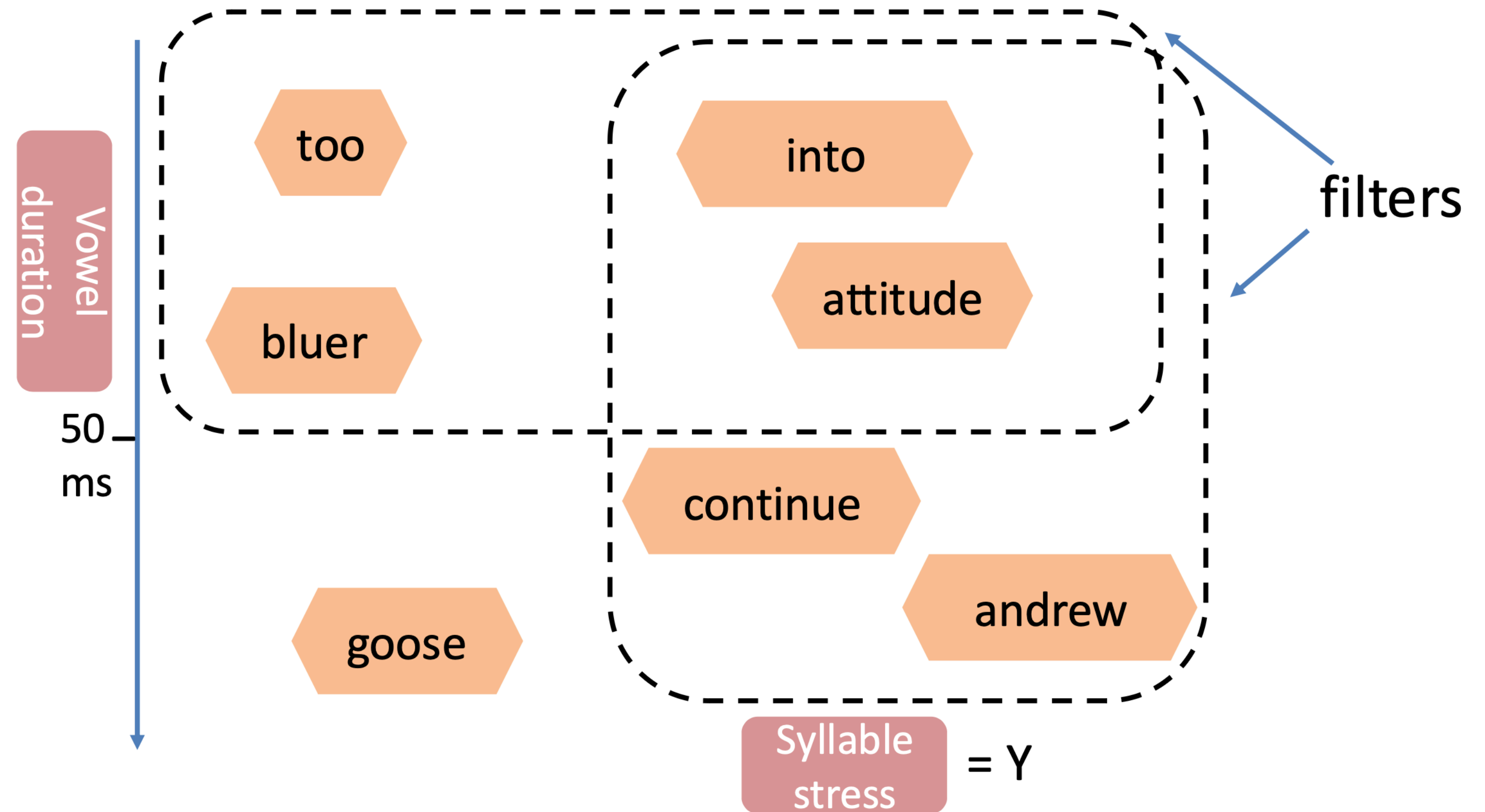
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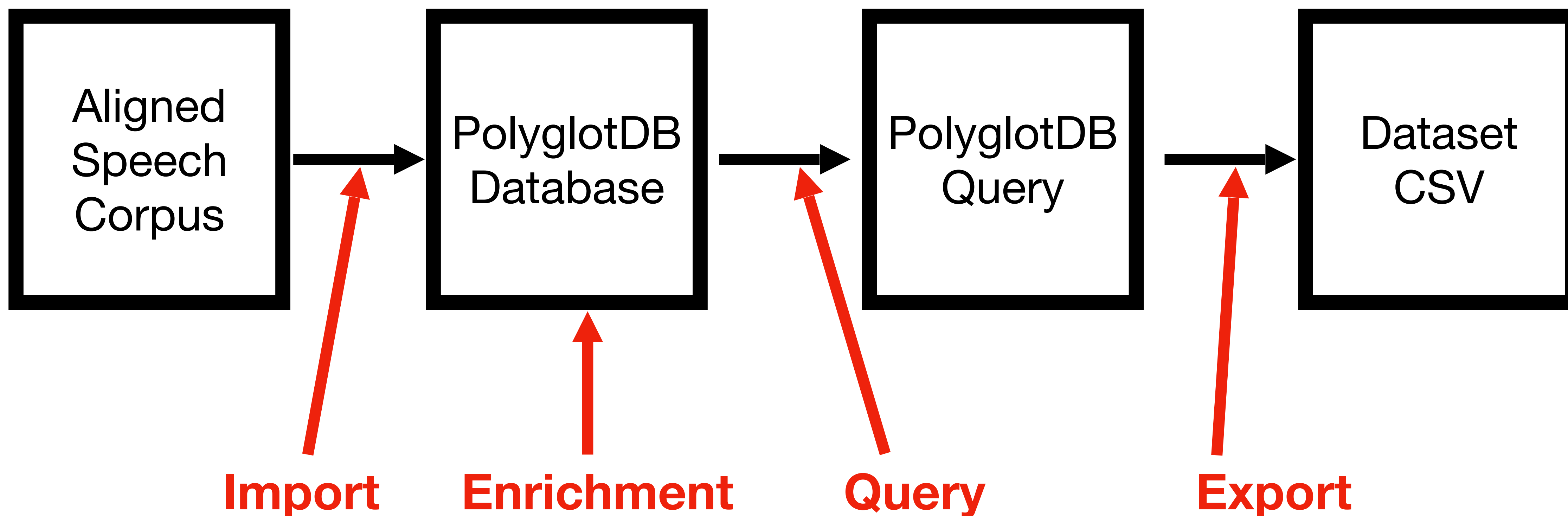
Query

- Select **unit of query**
 - Phones, syllables, utterances
- Provide **filters**
 - Stressed syllables, vowel segments, word-initial, etc

“Select /u:/ tokens in stressed syllables and a vowel duration $\geq 50\text{ms}$ ”

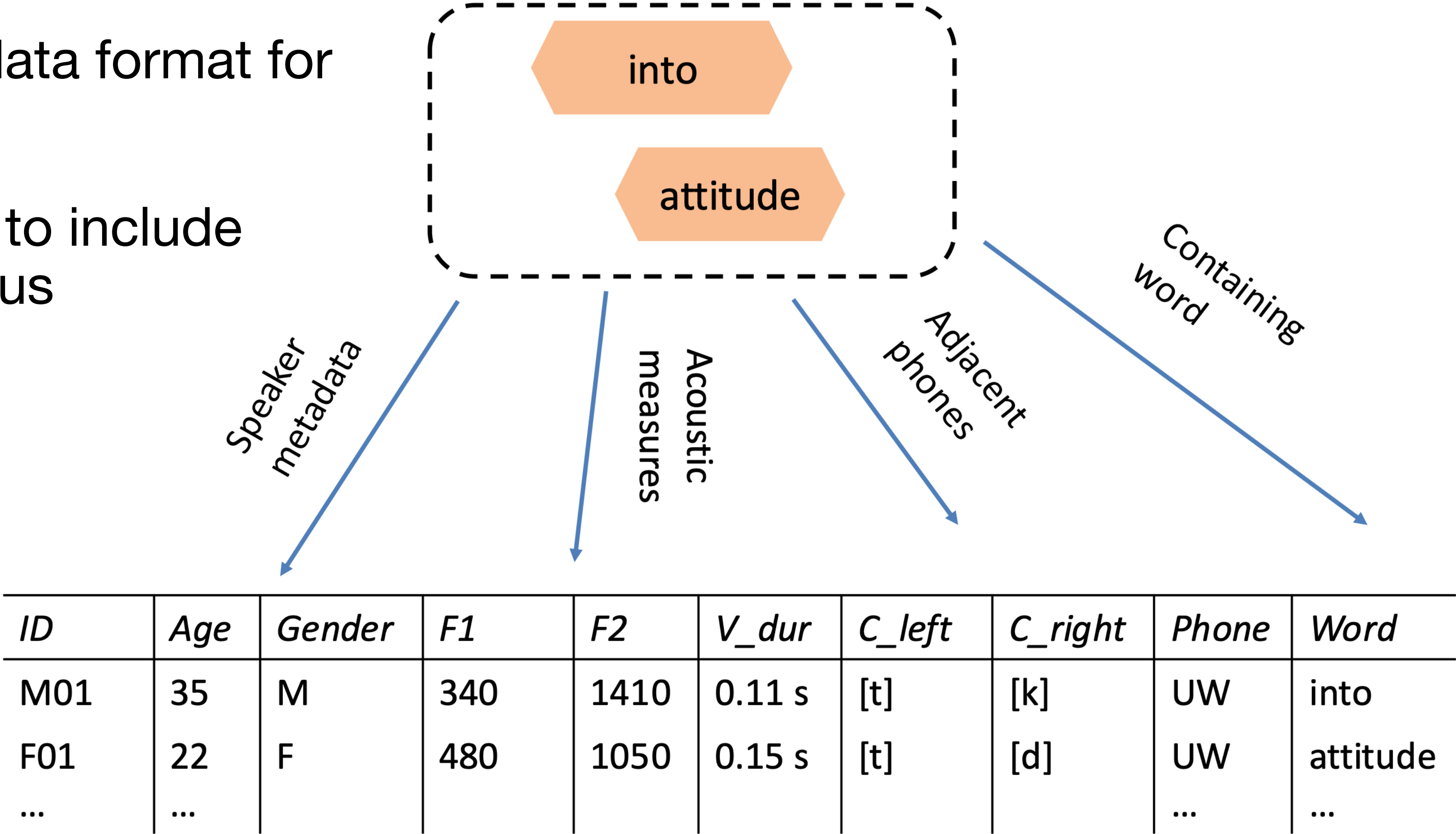


PolyglotDB workflow



Export

- **Write** query to data format for later analysis
- **Select** columns to include based on previous enrichments



Case Studies

Case Study 1



Toward “English” Phonetics: Variability in the Pre-consonantal Voicing Effect Across English Dialects and Speakers

James Tanner^{1*}, Morgan Sonderegger¹, Jane Stuart-Smith² and Josef Fruehwald³

¹ Department of Linguistics, McGill University, Montreal, QC, Canada, ² Glasgow University Laboratory of Phonetics, University of Glasgow, Glasgow, United Kingdom, ³ Department of Linguistics, University of Kentucky, Kentucky, KY, United States



English voicing effect (VE)

- Vowels longer before **voiced** consonants
 - bead > beat
- Varies in *size* across languages
 - English = ~1.6x; French = ~1.1x; Russian = 1.2x
- Studied only in **lab speech** (often single words!)
- Dialect-specific vowel patterning
 - SVLR
 - AAE

Research question

- *How robust is the Voicing Effect:*
 - *across dialects;*
 - *in spontaneous speech?*
- **Compare VE across different English dialects**

SPeech Across Dialects of English (SPADE) Project



- 66 corpora of North American & British Isles English speech
- Mixture of speech styles (spontaneous, interviews, reading tasks)
- 10,086 speakers, ~2400 hours
- Force-aligned with different aligners (MFA, FAVE, LaBB-CAT, MAUS)

Methodology

- Import each corpus using respective aligner parser
 - Import MFA-aligned corpora with PolyglotDB MFA parser, etc

Methodology

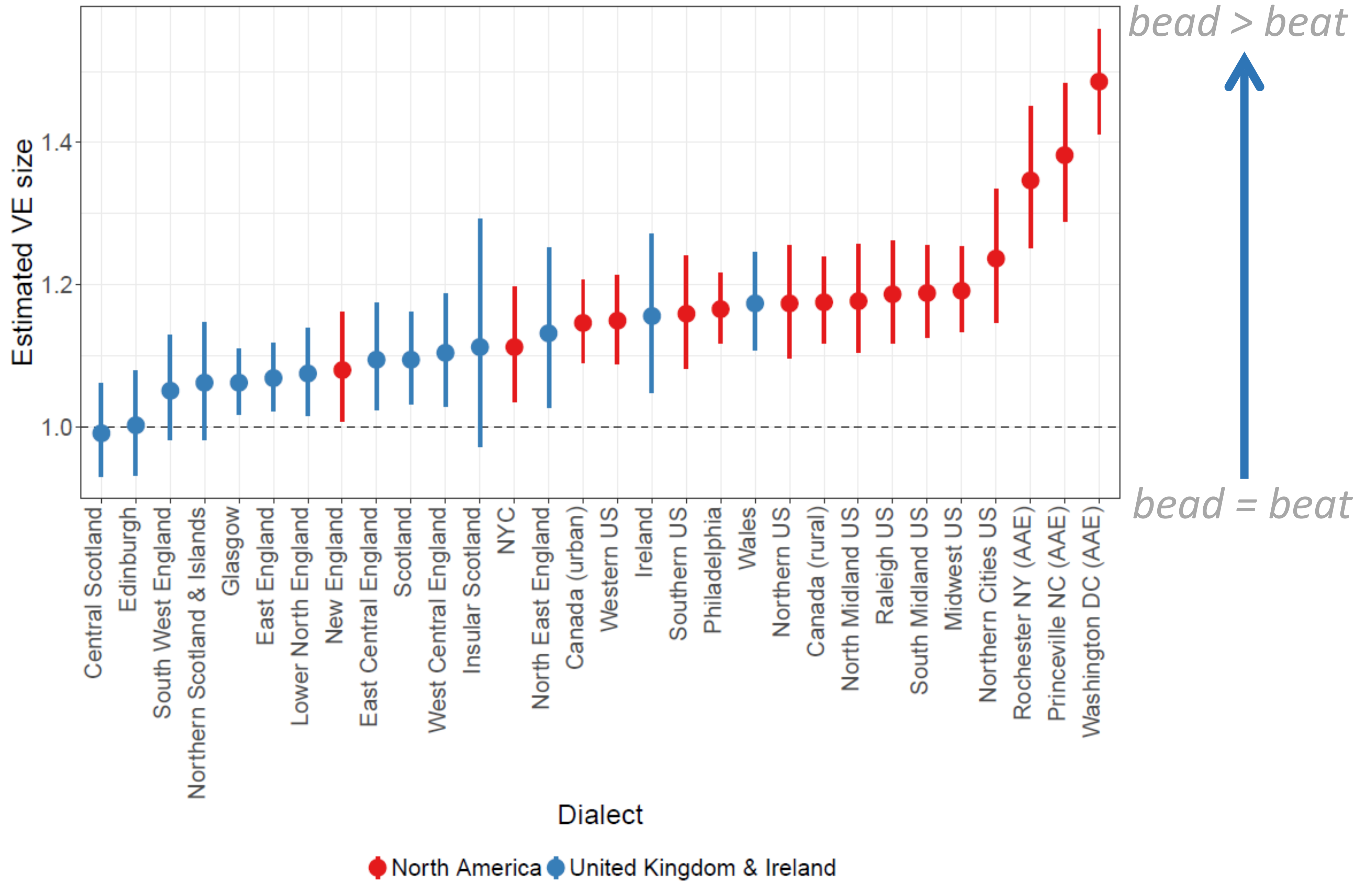
- Import each corpus using respective aligner parser
 - Import MFA-aligned corpora with PolyglotDB MFA parser, etc
- Perform **the same** enrichment for each corpus:
 - Encode utterances (150ms silence delimiter)
 - Encode syllabic structure
- Extract acoustic measure (vowel duration) along with other information (following consonant, dialect/speaker information)

Methodology

- Use 15 SPADE corpora reflecting NA/British Isles spontaneous speech:
 - ~230,000 vowel tokens
 - ~2,000 speakers
 - 30 NA/British regional dialects
- Fit Bayesian regression model of vowel duration, as function of:
 - Following C voicing
 - Following C POA
 - Vowel height
 - Speech rate
 - Dialect

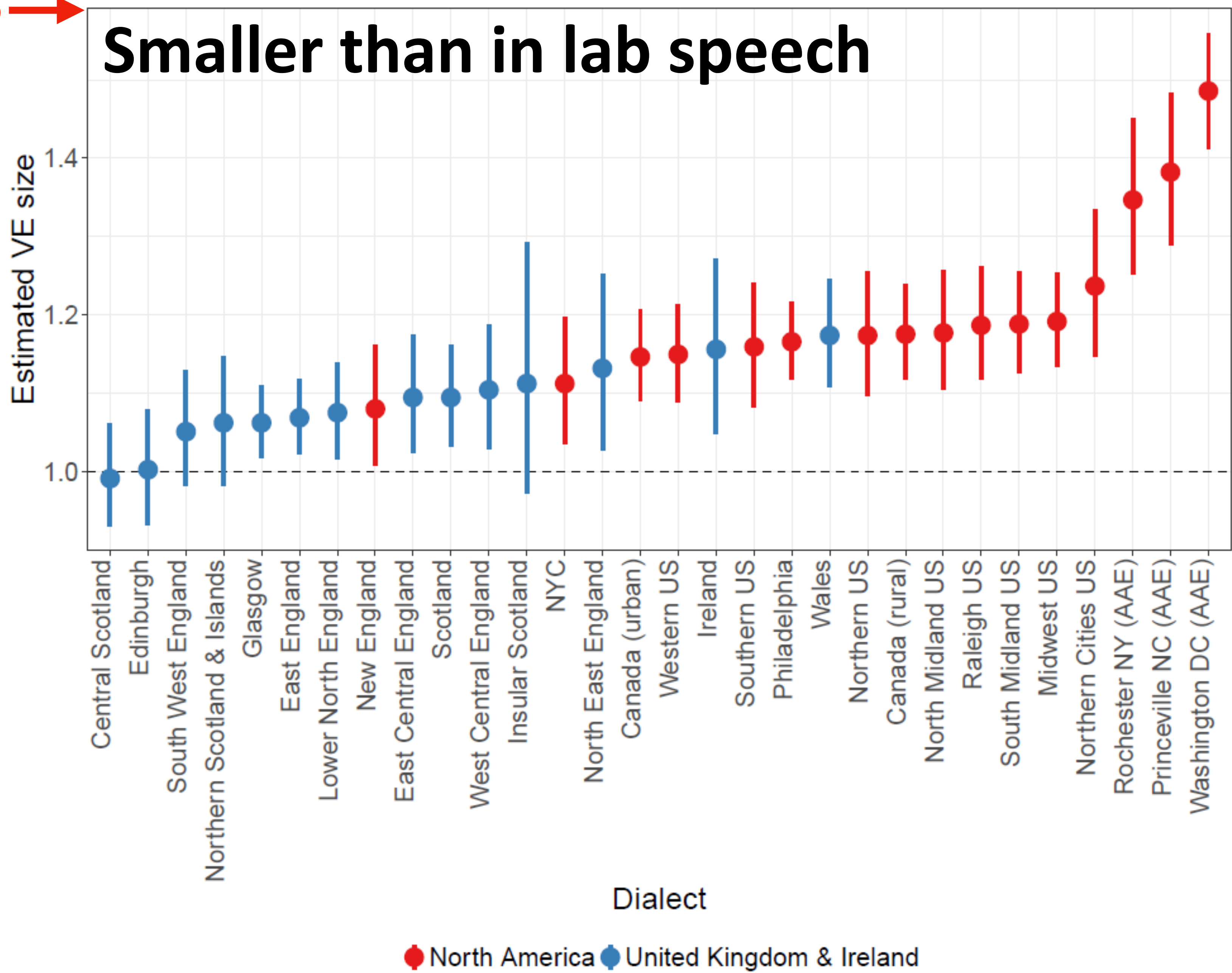
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Lab speech ~1.6 →

Smaller than in lab speech



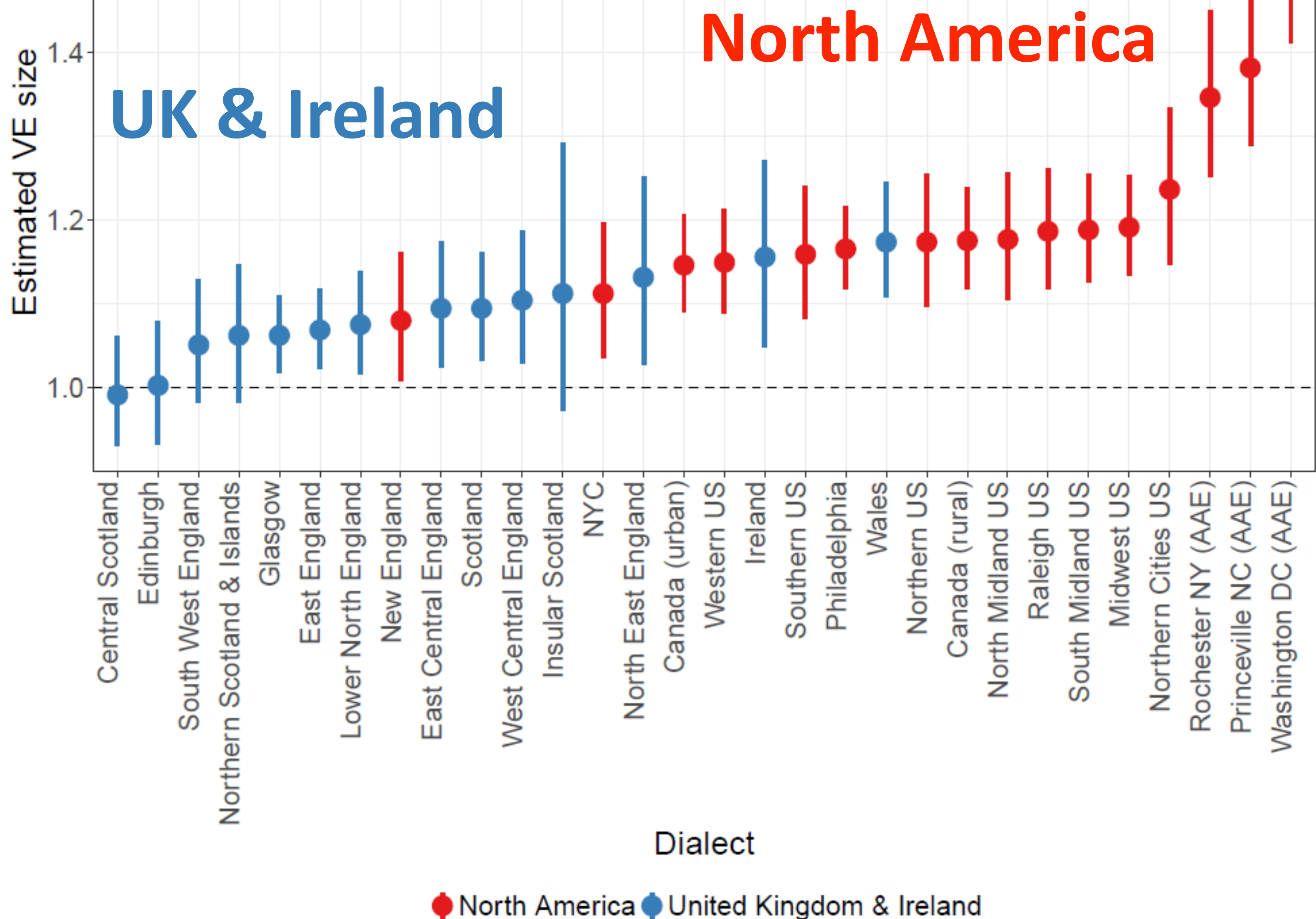
bead > beat



bead = beat

Lab speech ~1.6 →

NA Englishes larger than UK&I



bead > beat

bead = beat

Case Study 2

Interspeech 2024
1-5 September 2024, Kos, Greece



Exploring the anatomy of articulation rate in spontaneous English speech: relationships between utterance length effects and social factors

*James Tanner¹, Morgan Sonderegger², Jane Stuart-Smith¹, Tyler Kendall³, Jeff Mielke⁴, Robin
Dodsworth⁴, Erik Thomas⁴*

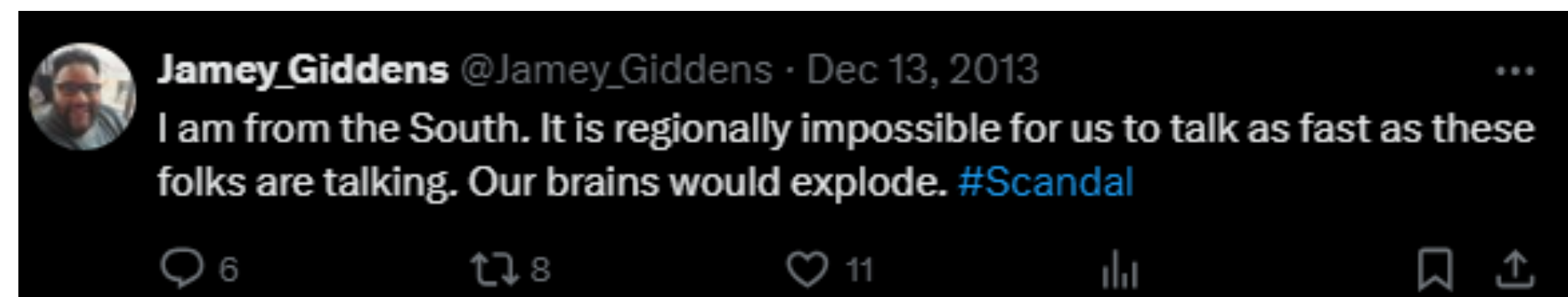
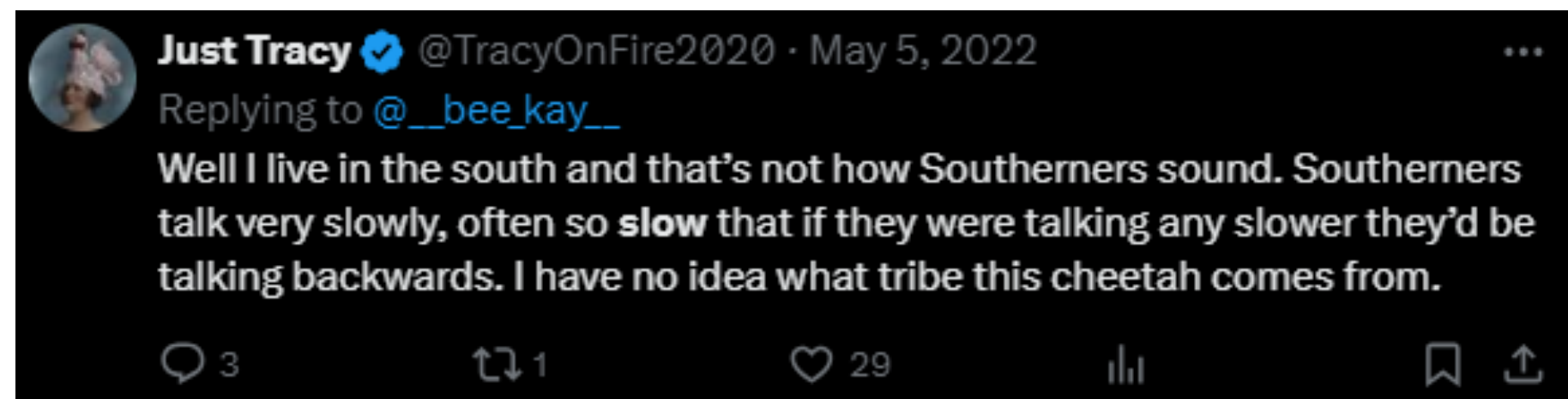
¹University of Glasgow, United Kingdom ²McGill University, Canada
³University of Oregon, United States ⁴North Carolina State University, United States

Now submitted to *Journal of Phonetics*

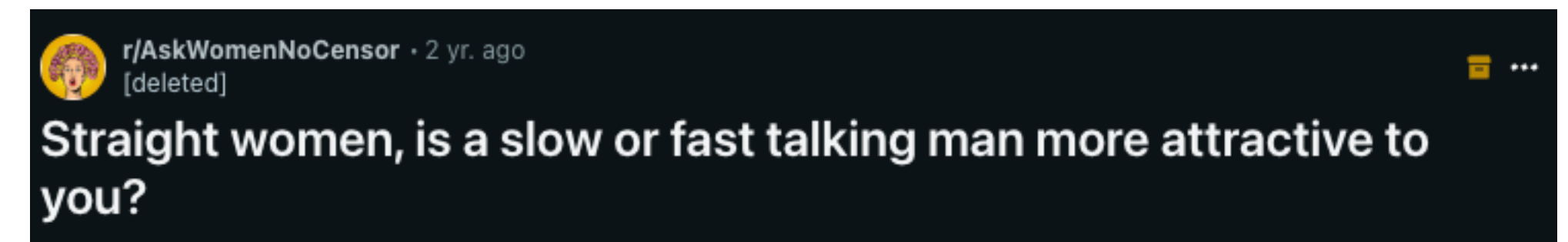
English speech rate

- Strong social stereotypes associated with how fast/slow people speak

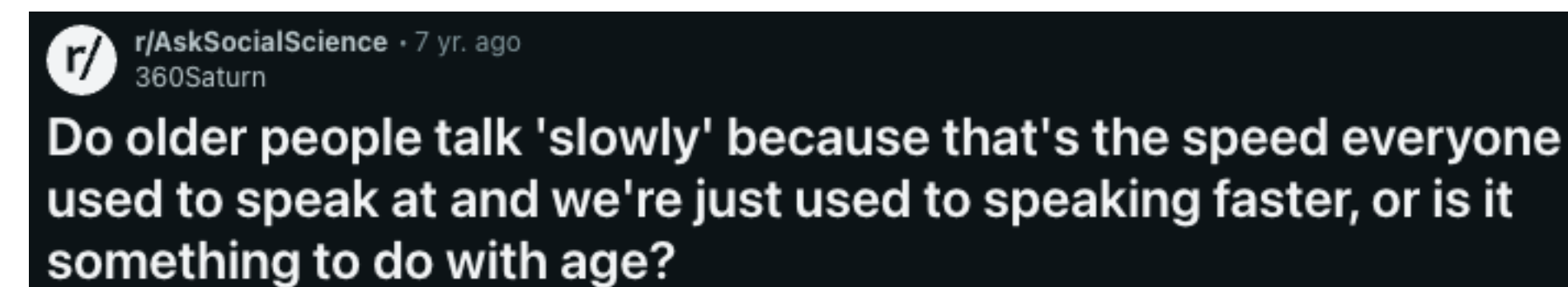
Dialect



Gender



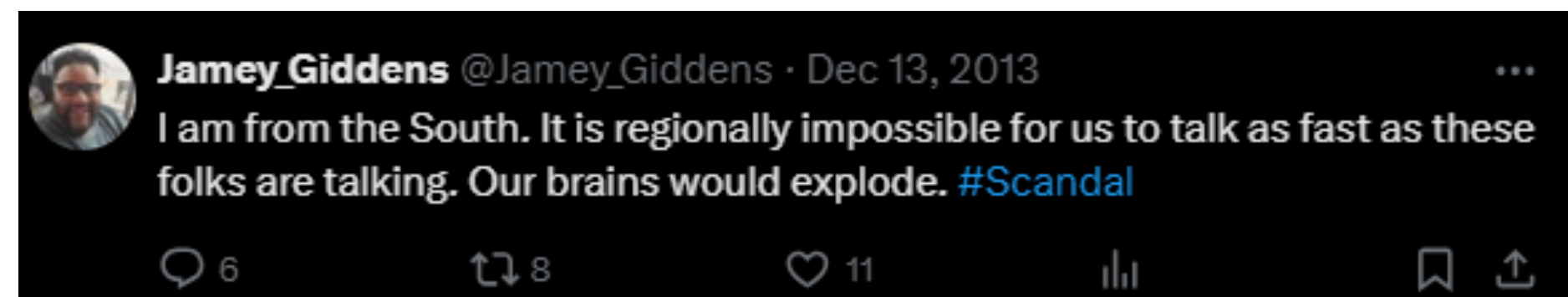
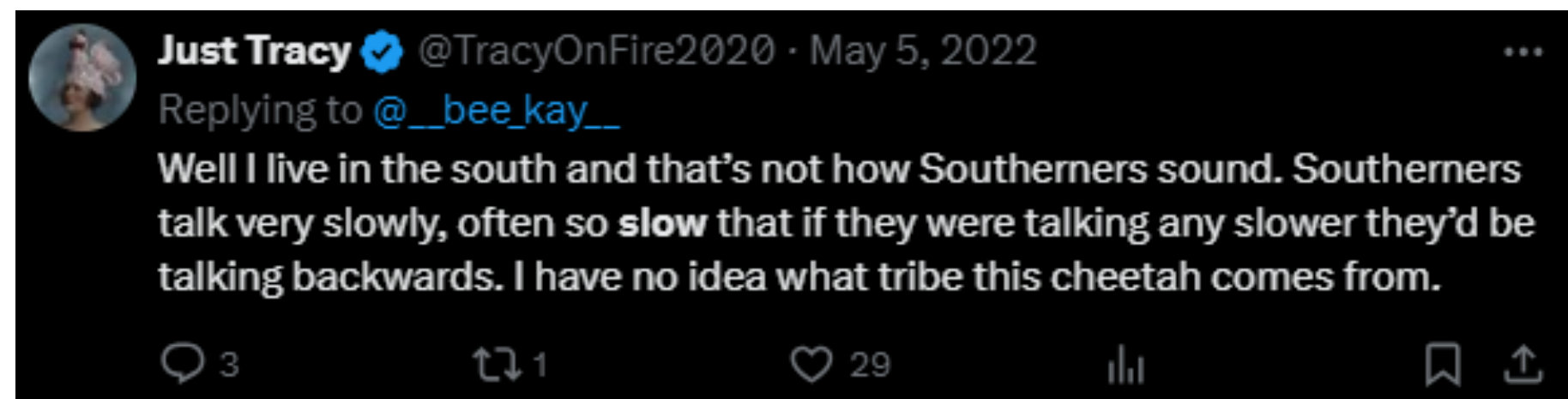
Age



English speech rate

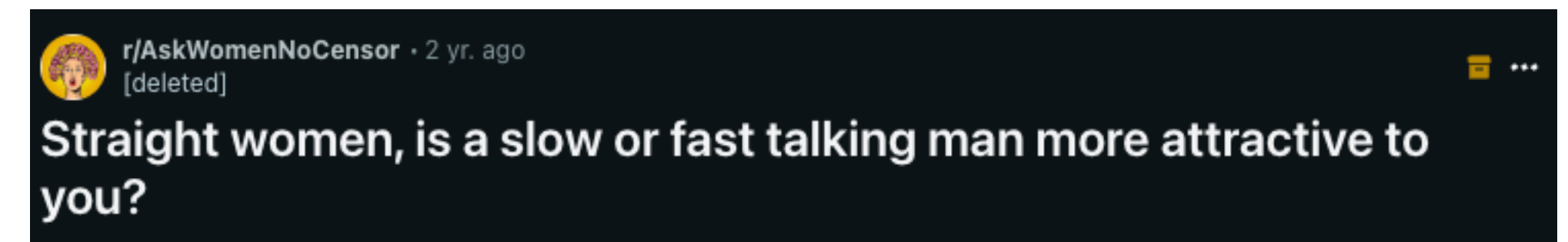
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Dialect

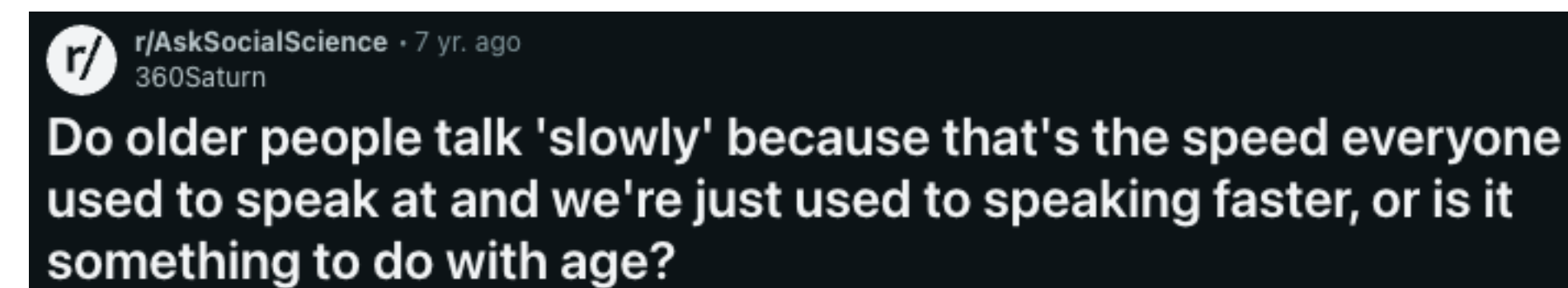


How big are these differences in reality?

Gender



Age



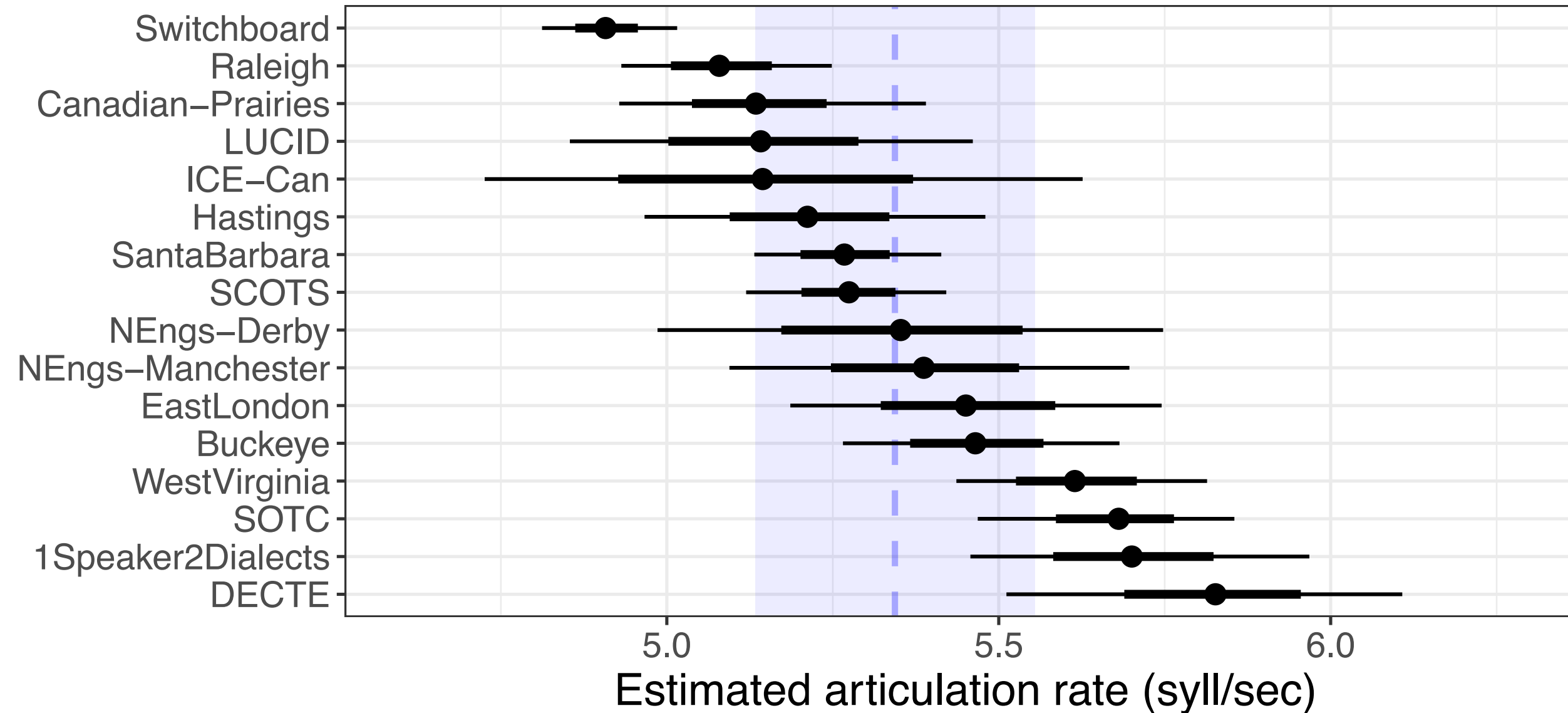
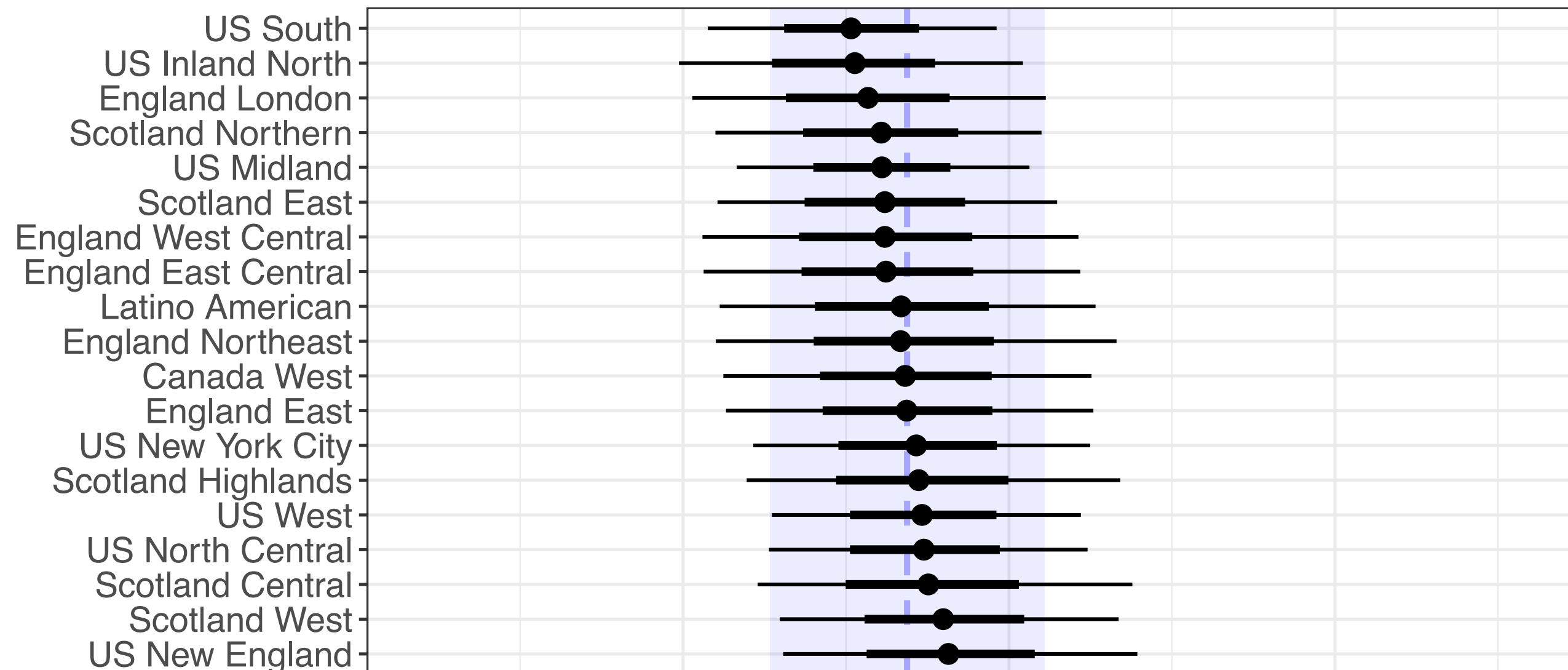
Methodology

- Encode utterance and syllabic information for each corpus
- Encode speech rate (syllables/sec) in PolyglotDB
- Model differences *across* sociolinguistic factors (dialects, genders)

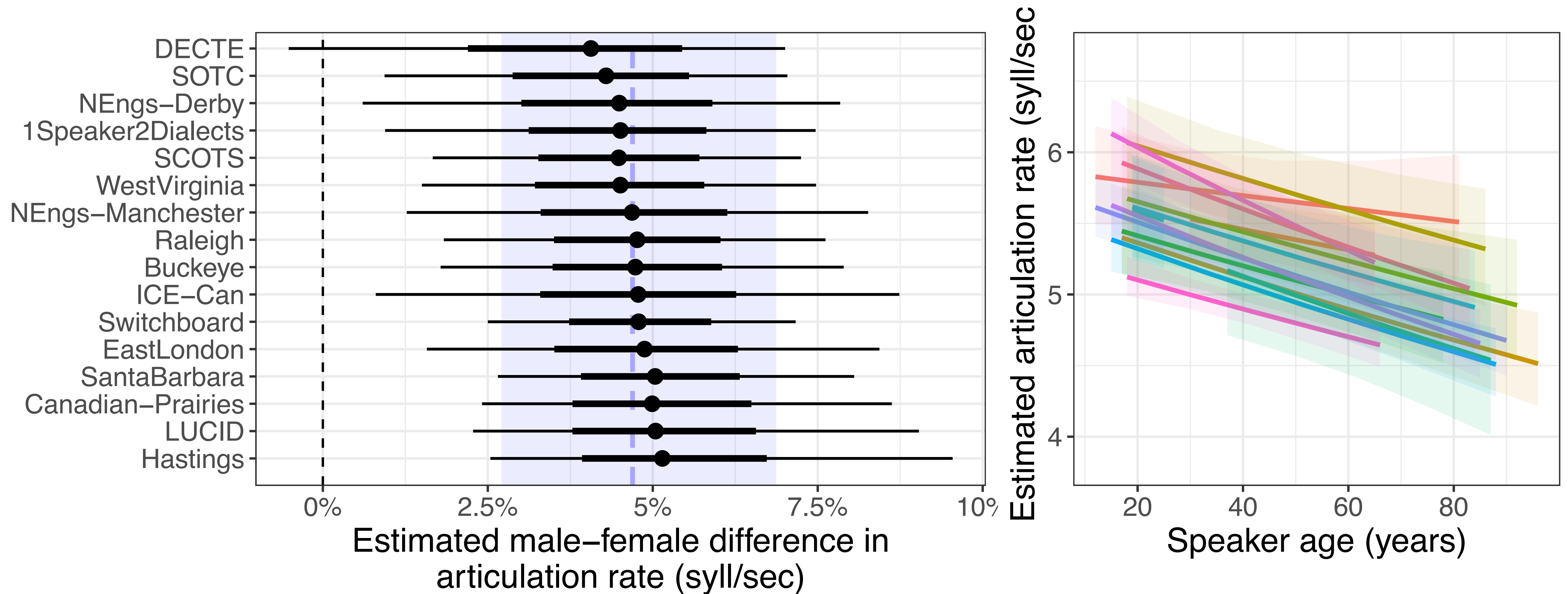


This study:
16 corpora =
19 British/North American
English dialects

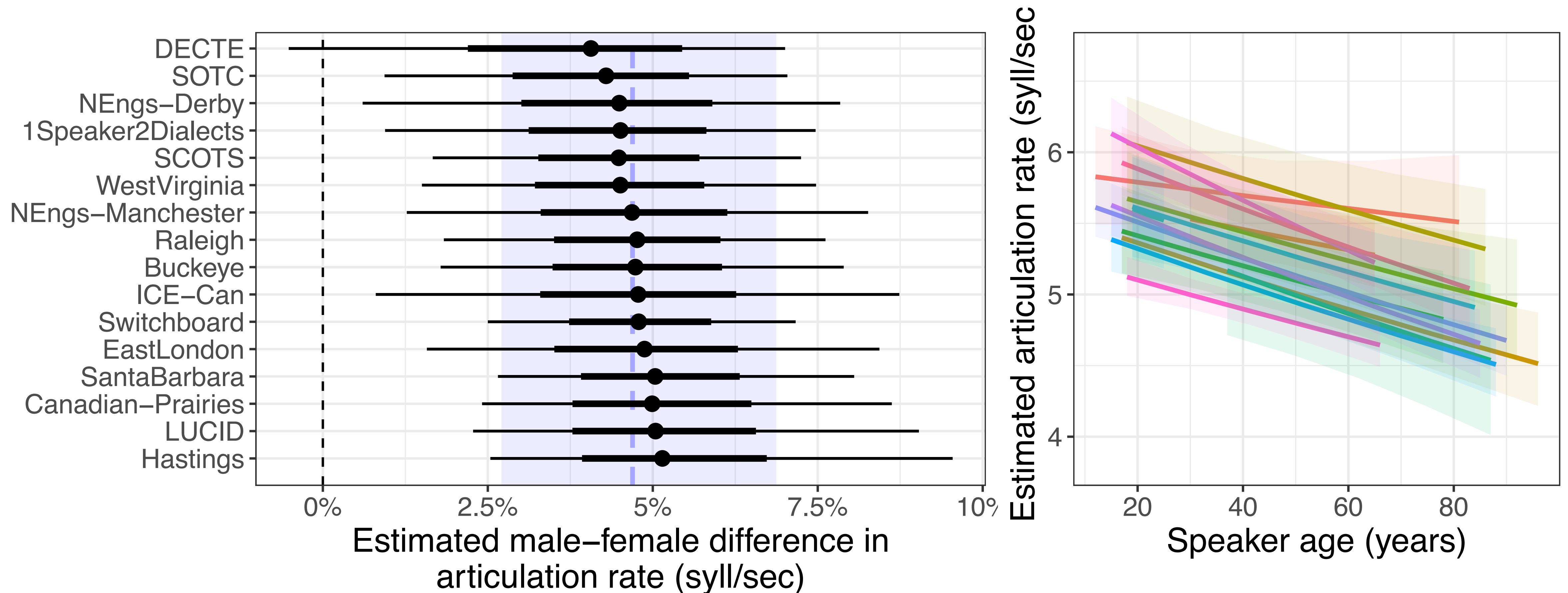
SPADE
SPeech Across Dialects of English



- More variation in speech rate across specific datasets (corpora) than across dialects!
- Suggests speech rate variation more tied to speech style



- Male speakers speak ~5% faster across corpora
- ~15% difference between youngest and oldest speakers



- Male speakers speak ~5% faster across corpora
- ~15% difference between youngest and oldest speakers

Social effects present but small

Thank you!

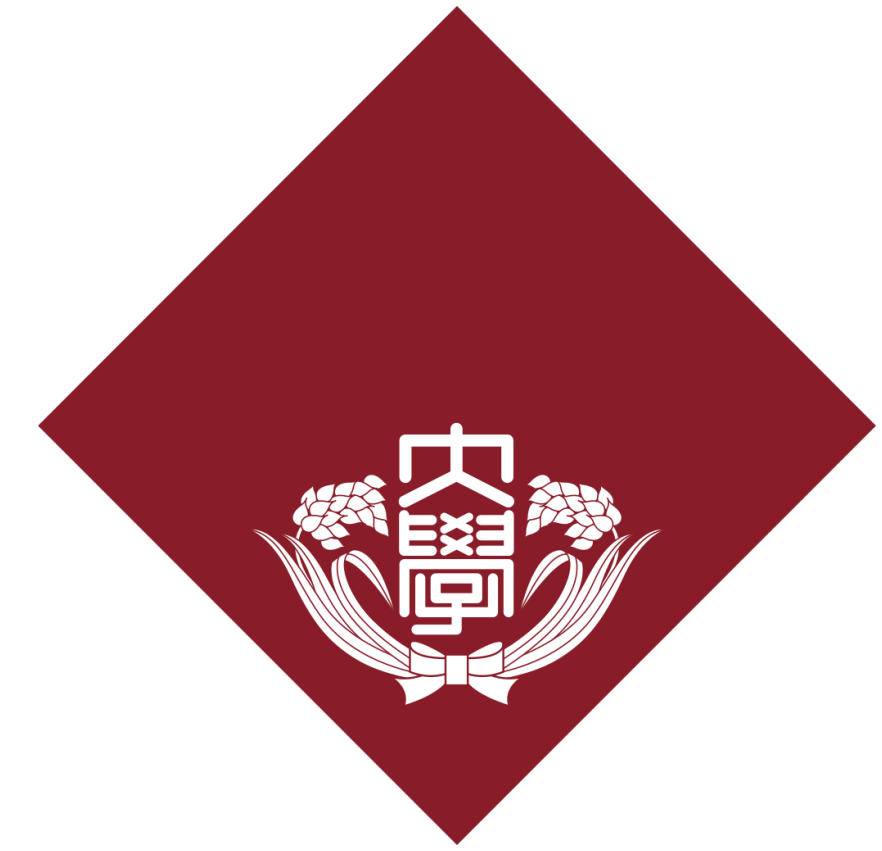
Seunghun Lee

Yasuki Shinohara

Michael McAuliffe

Morgan Sonderegger

Jane Stuart-Smith



Now it's your turn!

- After the break, we will explore how to use PolyglotDB:
- Going through some simple tutorials using Colab
- Explore applying PolyglotDB with your own data

<https://polyglotdb.readthedocs.io/en/latest/>

Link to Colab

<https://bit.ly/3Lck9TW>

